EDA and IDF-modeling

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Preperation

```r
if (!requireNamespace("pacman", quietly = TRUE)) {
  install.packages("pacman")
}
library(pacman)

# Load necessary libraries
p_load(caret)
p_load(tidyverse)
p_load(tidymodels)
p_load(tidytext)
p_load(textrecipes)
p_load(Matrix)
p_load(wordcloud)
p_load(tm)

p_load(stopwords)
p_load(RColorBrewer)

p_load(glmnet)
p_load(pROC)
```

```r
# ---------------------------------------------------------
# 1. Data Preparation
# ---------------------------------------------------------

df <- read_csv("../../data/processed/glassdoor_reviews_cleaned_v2.csv")
```

```
Rows: 1000 Columns: 32
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (12): employee_status, flags_business_outlook, flags_ceo_approval, flag...
dbl  (16): review_id, count_helpful, count_unhelpful, employee_job_end_year,...
lgl   (3): career_missing, employee_job_quit, has_review_text
date  (1): rating_date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# df <- read_csv("data/processed/glassdoor_reviews_cleaned_v2.csv")

cols <- c('summary', 'advice_to_management', 'review_pros', 'review_cons')
encode_cols <- c('covid_related_review', 'flags_business_outlook', 'flags_ceo_approval',
                 'flags_recommend_frend', 'rating_culture_values',
                 'rating_diversity_inclusion', 'rating_overall', 'rating_work_life',
                 'rating_compensation_benefits', 'rating_senior_leadership',
                 'rating_career_opportunities',
                 'has_review_text', 'career_missing')
df <- df %>%
  mutate(across(all_of(cols), ~replace_na(., "")))
# Define the columns
```

In the preliminary phase of the Exploratory Data Analysis (EDA), a word cloud was generated to visualize the most prominent terms within the comment dataset. To ensure the visualization reflected meaningful semantic patterns, the initial stopword list was expanded to exclude high-frequency terms that lacked domain-specific informational value.

```r
# 1. Define custom stopwords
# Equivalent to STOPWORDS.union(set([...]))
custom_stopwords <- c(stopwords("english"),
                      "bad", "good", "great", "will", "company",
                      "job", "work", "employee", "here",
                      "people", "employees")

# 2. Define the plotting function
plotWordCloudFromDF <- function(df, column_name) {

  # Extract text and collapse into one large string
  text_data <- paste(df[[column_name]], collapse = " ")
```

```r
  # Create a Corpus and clean the text
  docs <- Corpus(VectorSource(text_data))
  docs <- docs %>%
    tm_map(content_transformer(tolower)) %>%
    tm_map(removePunctuation) %>%
    tm_map(removeNumbers) %>%
    tm_map(removeWords, custom_stopwords) %>%
    tm_map(stripWhitespace)

  # Generate the Word Cloud
  cat(paste0("Generating word cloud for: ", column_name, "\n"))
  wordcloud(words = docs,
            max.words = 300,
            random.order = FALSE,
            rot.per = 0.35,            # Rotation percentage
            colors = brewer.pal(8, "Dark2"),
            scale = c(3, 0.5))        # Range of font sizes
}

# 3. Loop through columns


for (col in cols) {
  plotWordCloudFromDF(df, col)
}
```

```
Warning in tm_map.SimpleCorpus(., content_transformer(tolower)): transformation
drops documents

Warning in tm_map.SimpleCorpus(., removePunctuation): transformation drops
documents

Warning in tm_map.SimpleCorpus(., removeNumbers): transformation drops
documents

Warning in tm_map.SimpleCorpus(., removeWords, custom_stopwords):
transformation drops documents

Warning in tm_map.SimpleCorpus(., stripWhitespace): transformation drops
documents
```

```
Generating word cloud for: summary

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
supervisor could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
technology could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
unprofessional could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
values could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
zero could not be fit on page. It will not be plotted.
```

place
management
experience
environment
culture
worst
pay
terrible
toxic
horrible
leadership
career
avoid
stay
opportunities
best
senior
sales
okay
working
worklife
benefits
team
balance
manager
overall
better
potential
absolutely
business
opportunity
atmosphere
worth
little
hard
unless
dont
poor
away
nice
growth
start
workplace
overworked
organization
salary
family
excellent
dream
amazing
awesome
training
learning
youre
ceo
school
think
tech
high
never
pathetic
money
recommend
average
time
staff
broken
beware
positive
worked
individuals
bank
years
covid
industry
grow
growing
system
pace
interesting
wouldnt
department
analyst
wonderful
processes
plan
associate
leave
burnout
doesnt
change
fantastic
advisor
executive
disorganized
progression
way
option
small
safe
problems
mixed
conditions
costs
turnover
solid
due
fast
call
two
cool
far
fake
hours
life
decent
know
lots
foot
well
difficult
internal
direction
weak
upper
needs
door
lack
paid
just
many
want
much
under
move
loved
even
year
expected
outdated
integrity
thinking
seems
service
experienced
office
without
freshers
save
reviews
apply
lacking
corporate
one
day
look
marketing
consulting
product
health
nightmare
security
support
paid
practices
firm
agency
join
employer
managers
ruined
teamwork
treated
low
long
review
lot
half
's
ever
get
new
run
firstfun
like
awful
big
learn
little
hard
unless
never
ethics
broken
beware
positive
worked
individuals
think
school
tech
high
ceo
youre
learning
training
pace
interesting
wouldnt
department
analyst
wonderful
processes
plan
associate
leave
burnout
doesnt
change
fantastic
advisor
executive
disorganized
progression
way
option
family
salary
organization
small
safe
problems
mixed
conditions
costs
turnover
solid
due
fast
call
two
cool
far
fake
hours
life
decent
know
lots
foot
well
difficult
internal
direction
weak
upper
needs
door
lack
paid
just
many
want
much
under
move
loved
even
year
expected
outdated
integrity
thinking
seems
service
experienced
office
without
freshers
save
reviews
apply
lacking
corporate
one
day
look
marketing
consulting
product
health
nightmare
security
support
paid
practices
firm
agency
join
employer
managers
ruined
teamwork
treated
growing
system
grow
bank
years
covid
industry
worklife
benefits
recommend
average
time
staff
broken
beware
positive
worked
individuals
never
pathetic
money
ethics
high
tech
think
school

```
Warning in tm_map.SimpleCorpus(., removeNumbers): transformation drops
documents

Warning in tm_map.SimpleCorpus(., removeWords, custom_stopwords):
transformation drops documents

Warning in tm_map.SimpleCorpus(., stripWhitespace): transformation drops
documents

Generating word cloud for: advice_to_management

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
managers could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
turnover could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
years could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
actually could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
balance could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
clearly could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
continue could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
everyone could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
happy could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
hardworking could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
internal could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
learn could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
poor could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
positions could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
product could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
single could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
things could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
upper could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
valuable could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
working could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
accountable could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
approach could not be fit on page. It will not be plotted.
```

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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
changing could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
communication could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
create could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
department could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
develop could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
empathy could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
expectations could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
experiences could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
genuinely could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
grow could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
highly could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
hours could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
incentives could not be fit on page. It will not be plotted.
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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
interest could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
invest could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
learning could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
maintain could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
making could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
necessary could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, : one
could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
paying could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
perform could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
personal could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
playing could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
policies could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
politics could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
positive could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
practices could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
problems could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
project could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
projects could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
promote could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
properly could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
recognize could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
results could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
solely could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
someone could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
stores could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
supervisors could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
train could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
worklife could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
around could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
aspect could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
banks could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
base could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
bringing could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
broken could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
building could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
businesses could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
challenges could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
commission could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
compensation could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
conduct could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
consultants could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
contractors could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
contributions could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
days could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
despite could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
difference could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
different could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
difficult could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
done could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
drivers could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
encourage could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
enforce could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
engage could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
enough could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
environment could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
evaluations could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
formal could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
google could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
hires could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
horrible could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
however could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
humans could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
ignoring could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
improvement could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
interns could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
know could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
little could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
longer could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
looking could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
lowerlevel could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
made could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
members could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
model could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
morale could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
moving could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
much could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
needed could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
nothing could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
offer could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
operations could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
past could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
pressure could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
prevent could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
program could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
promoted could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
promoting could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
promotions could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, : put
could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
raises could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
really could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
recognition could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
reps could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
resources could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
responsibilities could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
retain could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
seriously could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
show could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
similar could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
start could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
store could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
stress could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
structure could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
students could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
talk could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
task could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
times could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
together could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
trust could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
trying could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
used could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
week could not be fit on page. It will not be plotted.

Warning in tm_map.SimpleCorpus(., content_transformer(tolower)): transformation
drops documents

Warning in tm_map.SimpleCorpus(., removePunctuation): transformation drops
documents

Warning in tm_map.SimpleCorpus(., removeNumbers): transformation drops
documents
```

```
Warning in tm_map.SimpleCorpus(., removeWords, custom_stopwords):
transformation drops documents

Warning in tm_map.SimpleCorpus(., stripWhitespace): transformation drops
documents

Generating word cloud for: review_pros

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
competitive could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
available could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
development could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
managers could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
professional could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
leadership could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
every could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
holidays could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
knowledge could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
provided could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
really could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
activities could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
around could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
atmosphere could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
building could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
enjoyed could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
friends could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
hardworking could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
members could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
overtime could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
package could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
security could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
unlimited could not be fit on page. It will not be plotted.
```

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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
week could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
wonderful could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
ability could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
advancement could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
allowing could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
areas could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
bonus could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
brand could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
cares could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
dedicated could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
events could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
genuinely could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
grow could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
kind could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
making could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
manager could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
much could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
overall could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
space could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
things could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
vacation could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
various could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
caring could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
companies could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
creative could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
direct could not be fit on page. It will not be plotted.
```

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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
family could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
impact could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
just could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
knowledgeable could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
learned could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
offering could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
paycheck could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
potential could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
resources could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
senior could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
software could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
students could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
technical could not be fit on page. It will not be plotted.
```

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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
technologies could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
annual could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
better could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
commission could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
compared could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
diverse could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
enjoyable could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
freedom could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
generous could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
global could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
help could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
individuals could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
leave could not be fit on page. It will not be plotted.
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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
lunch could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
meet could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
move could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
multiple could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
name could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
often could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
progression could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
provides could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
significant could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
stable could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
tech could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
travel could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
workfromhome could not be fit on page. It will not be plotted.
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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
year could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
awesome could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
challenging could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
community could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
customer could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
effort could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
especially could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
everything could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
fantastic could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
fulltime could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
group could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
international could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
know could not be fit on page. It will not be plotted.
```

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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
leaders could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
market could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
mission could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
offered could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
pandemic could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
pension could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
personal could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
plenty could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
position could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
pretty could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
providing could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
relaxed could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
rewarding could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
school could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
teamwork could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
thing could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
truly could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
allows could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
cafeteria could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
client could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
diversity could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
employer could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
engineering could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
entrylevel could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
extremely could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
fastpaced could not be fit on page. It will not be plotted.
```

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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
financial could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
gain could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
healthcare could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
inclusive could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
leading could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
long could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
match could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
meaningful could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
medical could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
meetings could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
minimal could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
model could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
money could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
never could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
ones could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
onsite could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
part could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
quickly could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
responsibilities could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
resume could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
reviews could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
seeking could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
sick could not be fit on page. It will not be plotted.
```

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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
size could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
solid could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
stay could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
still could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
success could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
system could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, : top
could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
weekends could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
willing could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
among could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
approach could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
encouraged could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
felt could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
patient could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
social could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
teammates could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
testing could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
youre could not be fit on page. It will not be plotted.

Warning in tm_map.SimpleCorpus(., content_transformer(tolower)): transformation
drops documents

Warning in tm_map.SimpleCorpus(., removePunctuation): transformation drops
documents

Warning in tm_map.SimpleCorpus(., removeNumbers): transformation drops
documents

Warning in tm_map.SimpleCorpus(., removeWords, custom_stopwords):
transformation drops documents

Warning in tm_map.SimpleCorpus(., stripWhitespace): transformation drops
documents

Generating word cloud for: review_cons

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
micromanagement could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
process could not be fit on page. It will not be plotted.
```

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Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
compensation could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
development could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
overworked could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
processes could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
unprofessional could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
departments could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
doesnt could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
internal could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
fired could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
higher could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
horrible could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
nonexistent could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
nothing could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
practices could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
requiring could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
reviews could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
systems could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
take could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
complete could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
corporate could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
done could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
everyone could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
getting could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
goals could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
help could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
instead could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
overtime could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
positions could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
salaries could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
shift could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
told could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
average could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
experienced could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
hiring could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
hour could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
inconsistent could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
individuals could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
mental could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
numbers could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
outdated could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
problems could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
rules could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
significant could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
store could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, : use
could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
worst could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
companies could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
decisions could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
direction could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
diversity could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
excessive could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
feedback could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
incompetent could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
leaving could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
makes could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
members could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
negative could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
promises could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
slow could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
stay could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
system could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
towards could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
unless could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
almost could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
calls could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
common could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
last could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
layoffs could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
longterm could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, : may
could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
political could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
products could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
progression could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
promotion could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
provided could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
requires could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
rude could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
small could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
workers could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
avoid could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
competitive could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
disorganized could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
enough could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
give could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
ideas could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
lacking could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
service could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
technical could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
technology could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
underpaid could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
unrealistic could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
among could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
breaks could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
colleagues could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
family could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
frequent could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
hires could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
including could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
industry could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
keep could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
knowledge could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
notice could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
product could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
shifts could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
students could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
used could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
want could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
well could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
claims could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
companys could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
expect could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
ineffective could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
meeting could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
minimal could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
mistakes could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
owners could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
policies could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
promised could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
promoted could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
quickly could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
rate could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
resources could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
right could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
success could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
takes could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
bullying could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
client could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
holidays could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, : old
could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
period could not be fit on page. It will not be plotted.
```

```
Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
program could not be fit on page. It will not be plotted.

Warning in wordcloud(words = docs, max.words = 300, random.order = FALSE, :
revenue could not be fit on page. It will not be plotted.
```

The above is the data preperation step, mix up TF-IDF, dummy variable and dataset split.

```
# Split Data (equivalent to train_test_split)
set.seed(42)
df2 = df %>%
  subset(select = -c(rating_date, employee_status, company_name,
  employee_location, employee_job_title, sector, employee_job_end_year,
  review_id))


# Build the Preprocessing Recipe (Equivalent to ColumnTransformer + TfidfVectorizer)
rec <- recipe(employee_length + employee_job_quit ~ ., data = df2)  %>%
  step_mutate(across(all_of(encode_cols), as.factor)) %>%
  # Handle Numeric/Categorical preprocessing
  step_novel(all_of(encode_cols)) %>%
  step_dummy(all_of(encode_cols), one_hot = FALSE) %>%

  # Text Processing for 'summary' (max_features = 200)
  step_tokenize(summary) %>%
  step_stopwords(summary) %>%
  step_ngram(summary, min_num_tokens = 1, num_tokens = 3) %>%
  step_tokenfilter(summary, max_tokens = 200, min_times = 5) %>%
  step_tfidf(summary, prefix = "") %>%

  # Text Processing for 'advice_to_management' (max_features = 200)
  step_tokenize(advice_to_management) %>%
```

```r
  step_stopwords(advice_to_management) %>%
  step_ngram(advice_to_management, min_num_tokens = 1, num_tokens = 3) %>%
  step_tokenfilter(advice_to_management, max_tokens = 200, min_times = 5) %>%
  step_tfidf(advice_to_management, prefix = "") %>%

  # Text Processing for 'review_pros' (max_features = 300)
  step_tokenize(review_pros) %>%
  step_stopwords(review_pros) %>%
  step_ngram(review_pros, min_num_tokens = 1, num_tokens = 3) %>%
  step_tokenfilter(review_pros, max_tokens = 300, min_times = 5) %>%
  step_tfidf(review_pros, prefix = "") %>%

  # Text Processing for 'review_cons' (max_features = 400)
  step_tokenize(review_cons) %>%
  step_stopwords(review_cons) %>%
  step_ngram(review_cons, min_num_tokens = 1, num_tokens = 3) %>%
  step_tokenfilter(review_cons, max_tokens = 400, min_times = 5) %>%
  step_tfidf(review_cons, prefix = "") %>%
  step_zv(all_predictors()) %>%
  # Scaling (Equivalent to MaxAbsScaler)
  # R's step_normalize is Z-score; for MaxAbs we use a custom scaling or range
  step_mutate(across(where(is.numeric), ~replace_na(., 0))) %>%
  step_range(all_predictors(), min =0, max = 1)


set.seed(42)
data_split <- df2$employee_job_quit%>%
  createDataPartition(p = 0.8, list = F)
train_data <- df2[data_split, ]
test_data  <- df2[-data_split, ]

# Train the recipe on training data
prep_rec <- prep(rec, training = train_data)
```

```
Warning: ! There are new levels in `flags_ceo_approval`: NA.
i Consider using step_unknown() (`?recipes::step_unknown()`) before
  `step_dummy()` to handle missing values.

Warning: max_tokens was set to 200, but only 108 was available and selected.

Warning: max_tokens was set to 200, but only 106 was available and selected.
```

```r
# Transform the datasets (Equivalent to fit_transform / transform)
X_train_processed <- bake(prep_rec, new_data = train_data, all_predictors(), composition = "c
```

```
Warning: ! There are new levels in `flags_ceo_approval`: NA.
i Consider using step_unknown() (`?recipes::step_unknown()`) before
  `step_dummy()` to handle missing values.
```

```r
X_test_processed  <- bake(prep_rec, new_data = test_data, all_predictors(), composition = "dg

# Target vectors
yl_train <- train_data$employee_length
yq_train <- as.integer(train_data$employee_job_quit)

yl_test <- test_data$employee_length
yq_test <- as.integer(test_data$employee_job_quit)

# --- Data Structural Audit ---
cat("--- Data Structural Audit ---\n")
```

```
--- Data Structural Audit ---
```

```r
cat(sprintf("Total Combined Features: %d\n", ncol(X_train_processed)))
```

```
Total Combined Features: 958
```

```r
cat(sprintf("Total Samples: %d\n", nrow(X_train_processed)))
```

```
Total Samples: 800
```

```r
cat("First 30 Feature Names (Sample):\n")
```

```
First 30 Feature Names (Sample):
```

```r
print(colnames(X_train_processed)[1:30])
```

```
 [1] "count_helpful"                   "count_unhelpful"
 [3] "rating_year"                     "helpfulness_score"
 [5] "employee_job_start_year"         "covid_related_review_X1"
 [7] "flags_business_outlook_NEUTRAL"  "flags_business_outlook_POSITIVE"
 [9] "flags_ceo_approval_DISAPPROVE"   "flags_ceo_approval_NO_OPINION"
[11] "flags_recommend_frend_POSITIVE"  "rating_culture_values_X2"
[13] "rating_culture_values_X3"        "rating_culture_values_X4"
[15] "rating_culture_values_X5"        "rating_diversity_inclusion_X2"
[17] "rating_diversity_inclusion_X2.5" "rating_diversity_inclusion_X3"
[19] "rating_diversity_inclusion_X3.5" "rating_diversity_inclusion_X4"
[21] "rating_diversity_inclusion_X4.5" "rating_diversity_inclusion_X5"
[23] "rating_overall_X2"               "rating_overall_X3"
[25] "rating_overall_X4"               "rating_overall_X5"
[27] "rating_work_life_X2"             "rating_work_life_X3"
[29] "rating_work_life_X4"             "rating_work_life_X5"
```

word frequency

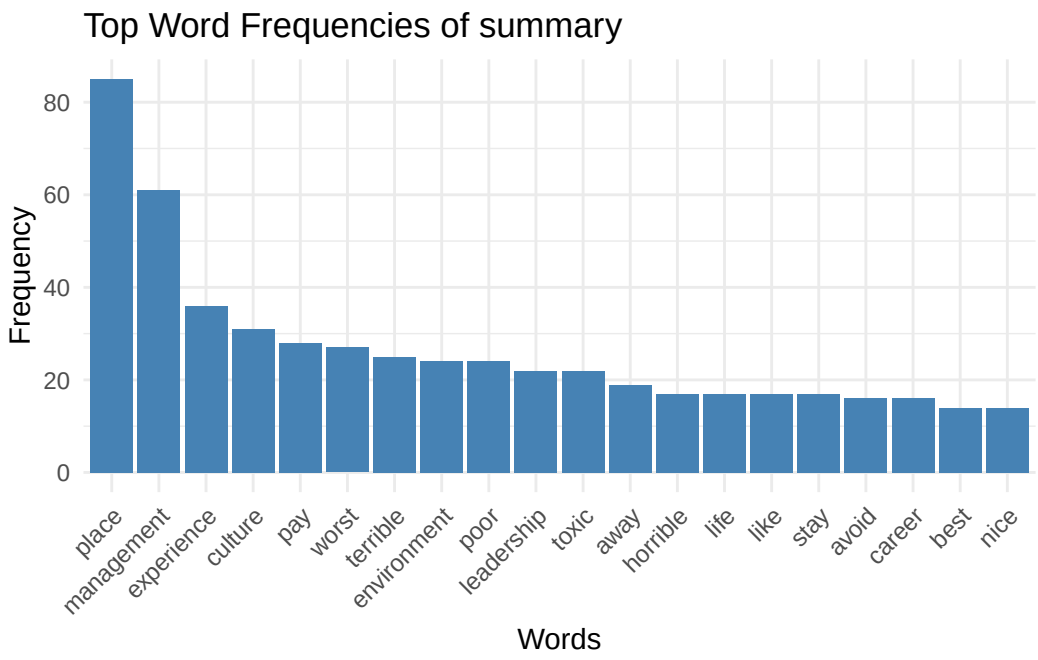
This step focus on how to choose optimal parameter for TF-IDF. We gonna plot the word frequency bar chart. The stopwords should be removed since meaningless, and also the low-frequency words for being noisy.

Since we have feature selection step in the further, we would rather import more words here.

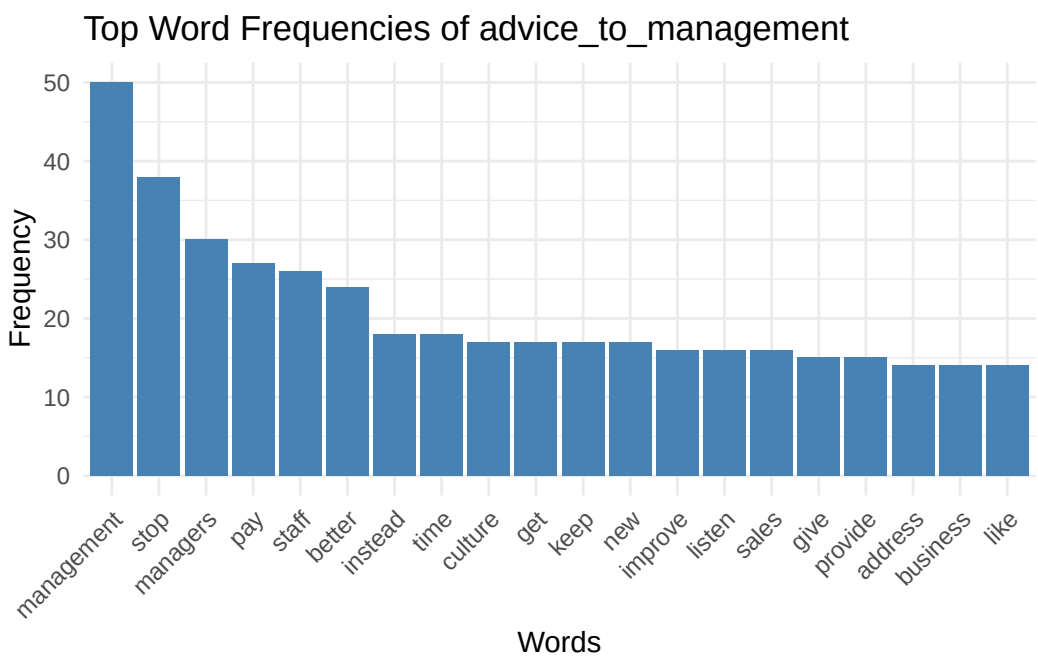
```r
# clean function
clean_words <- function(data, column_name) {

  # 1. convert to tidy (Tokenization & Cleaning)
  # unnest_tokens will transfer to lowercase and delete non-letter chars
  tidy_text <- data %>%
    select(all_of(column_name)) %>%
    unnest_tokens(word, !!sym(column_name))

  # total words
  wa <- tidy_text %>% distinct(word) %>% nrow()

  # 2. filter stopwords and low frequency
  # custom_stopwords is defined before
  cleaned_text <- tidy_text %>%
    filter(!word %in% custom_stopwords) %>%
    group_by(word) %>%
    filter(n() >= 5) %>% # drop words less than 5 times
```

```r
    ungroup()

  # word number after filter
  wb <- cleaned_text %>% distinct(word) %>% nrow()

  # print the statistics
  cat(sprintf("%s original words: %d  stopword and lowfreq removed: %d\n",
              column_name, wa, wb))

  # 3. calculate the top 20
  word_counts <- cleaned_text %>%
    count(word, sort = TRUE) %>%
    slice_head(n = 20)

  # 4. visualization
  p <- ggplot(word_counts, aes(x = reorder(word, -n), y = n)) +
    geom_col(fill = "steelblue") +
    labs(title = paste("Top Word Frequencies of", column_name),
         x = "Words", y = "Frequency") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))

  print(p)

  # return complete word frequency
  full_counts <- cleaned_text %>%
    count(word, sort = TRUE)

  return(full_counts)
}

results <- list()
for (col in cols) {
  results[[col]] <- clean_words(df, col)
}
```

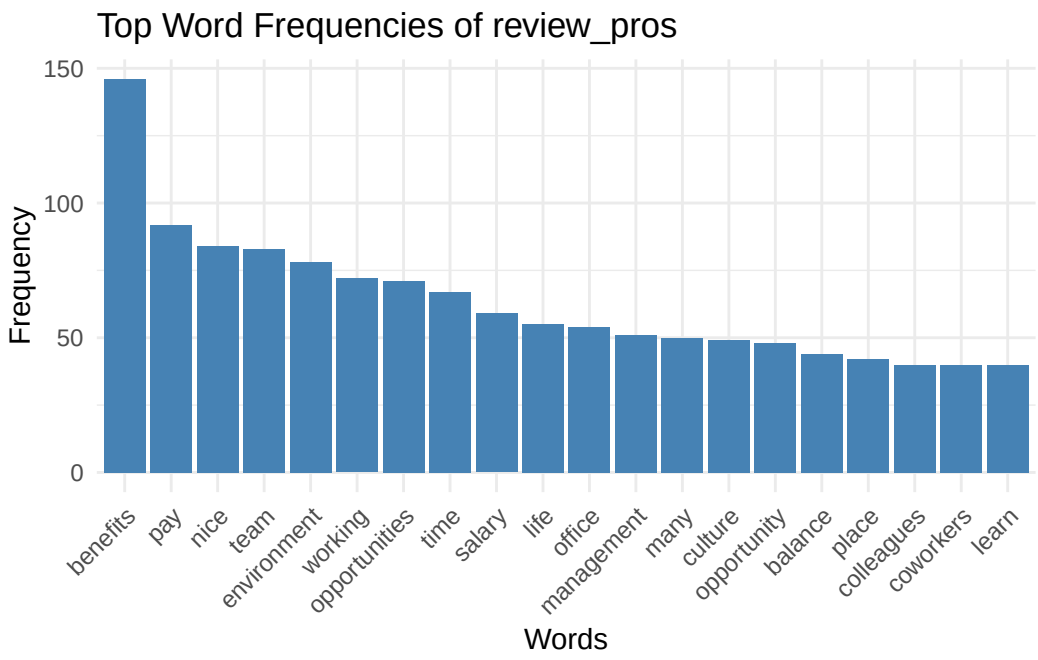
```
summary original words: 1279  stopword and lowfreq removed: 98
```



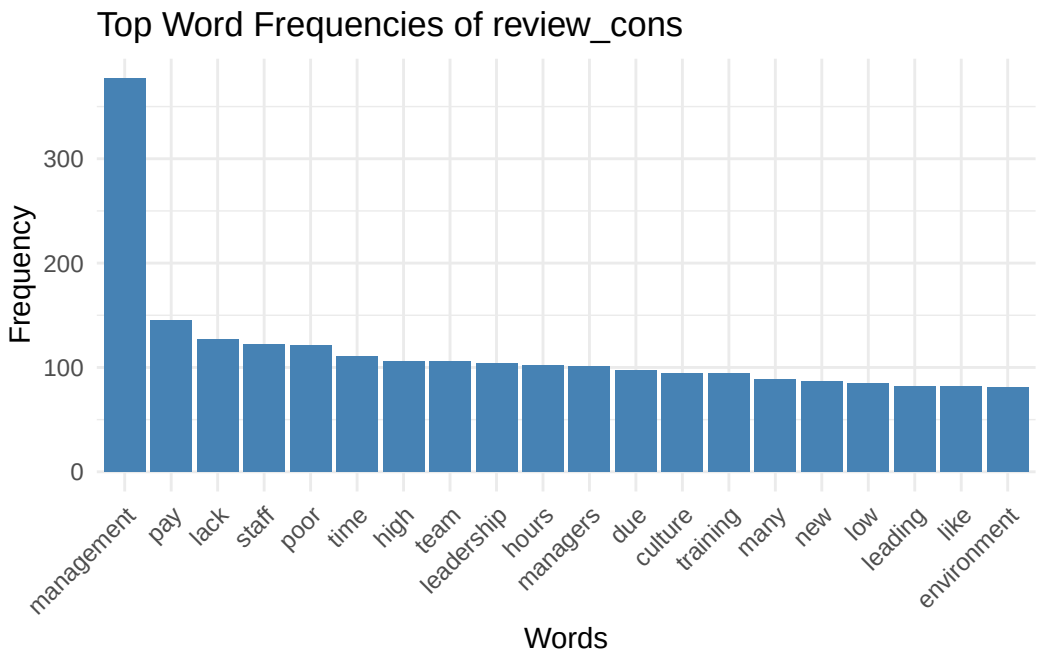

```
advice_to_management original words: 1645  stopword and lowfreq removed: 127
```




```
review_pros original words: 2592  stopword and lowfreq removed: 364
```




```
review_cons original words: 5291  stopword and lowfreq removed: 1026
```



Quit Model

This model focus on whether people quit their job, and what factor affect mostly in their decision of job-quitting.

Here we use Lasso-Logistics to select most important factor, Ridge-Logistics, and also Elastic-Logistics.

Lasso

```
tab <- table(yq_train)
weights <- ifelse(yq_train == 1,
                  sum(tab) / (2 * tab["1"]),
                  sum(tab) / (2 * tab["0"]))


lambda <- 10^seq(1,-2,length = 100)
model.quit.lasso <- cv.glmnet(X_train_processed, yq_train, family = binomial, alpha = 1, lamb
```

```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
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```

```
model.quit.lasso_coef <- predict(model.quit.lasso, s = model.quit.lasso$lambda.min, type = "c

model.quit.lasso_coefs <- as.matrix(model.quit.lasso_coef)
model.quit.lasso_vars <- data.frame(
  Variable = rownames(model.quit.lasso_coefs),
  Coefficient = as.numeric(model.quit.lasso_coefs)
) %>% filter(Coefficient != 0 & Variable != "(Intercept)") %>%
  arrange(Coefficient)
```

```r
model.quit.lasso_vars
```

```
                        Variable Coefficient
1              _review_cons_covid -2.51765033
2         covid_related_review_X1 -1.87597535
3               _review_pros_work -1.56647481
4        _review_pros_advancement -1.27183916
5              _review_cons_still -1.26020402
6            _review_pros_support -1.01404222
7              _review_pros_right -0.97102523
8        _summary_company_culture -0.79066271
9               _review_cons_lead -0.76941353
10               _review_cons_see -0.75713320
11              _review_pros_grow -0.63517547
12          _review_cons_requires -0.56852296
13          _review_pros_discount -0.53979000
14         _review_pros_coworkers -0.45637558
15       _review_pros_competitive -0.45533330
16             _review_pros_great -0.35623607
17               _summary_culture -0.25891275
18            _review_cons_issues -0.25642983
19    _advice_to_management_focus -0.14786895
20          _review_pros_provides -0.14739981
21            rating_work_life_X4 -0.11165693
22              _review_cons_well -0.07005663
23        employee_job_start_year -0.04596288
24                   count_helpful -0.04023575
25              rating_overall_X5 -0.03372454
26        _review_pros_place_work  0.23345853
27 rating_career_opportunities_X3  0.24569653
28  flags_ceo_approval_DISAPPROVE  0.32067360
29              _review_cons_long  0.90574593
30       _review_pros_environment  1.09356007
31           career_missing_TRUE.  1.18010892
```

```r
roc_obj <- roc(yq_train, predict(model.quit.lasso, newx = X_train_processed, type = "respons
```

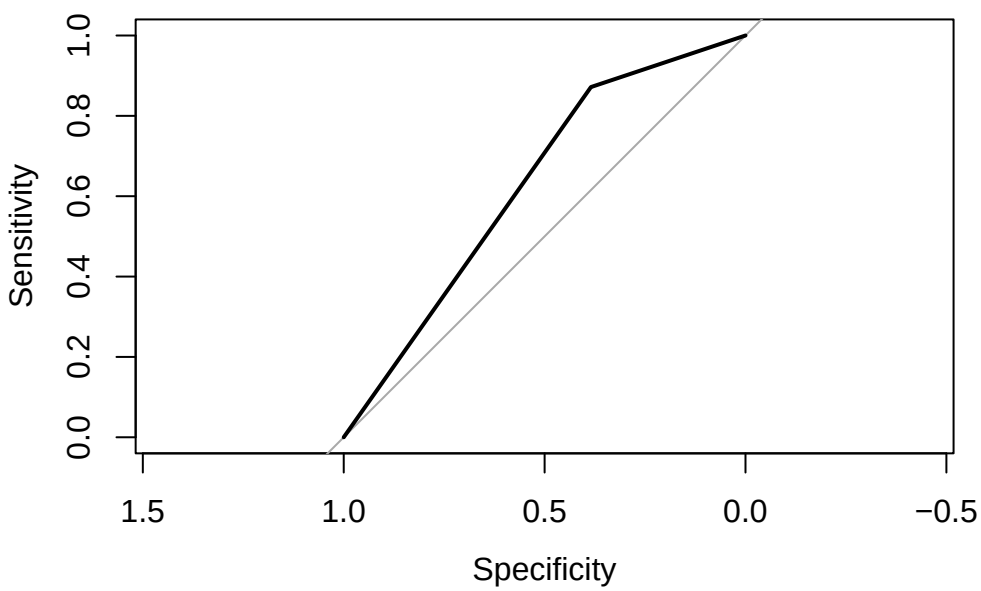
```
Setting levels: control = 0, case = 1
```

```
Warning in roc.default(yq_train, predict(model.quit.lasso, newx =
X_train_processed, : Deprecated use a matrix as predictor. Unexpected results
may be produced, please pass a numeric vector.

Setting direction: controls < cases
```

```
best_threshold.lasso <- coords(roc_obj, "best", ret = "threshold")[1,1]

yq_pred.lasso <- ifelse(predict(model.quit.lasso, newx = X_test_processed, type = "response")
log_roc_curve.lasso <- roc(yq_test, yq_pred.lasso)
```

```
Setting levels: control = 0, case = 1

Warning in roc.default(yq_test, yq_pred.lasso): Deprecated use a matrix as
predictor. Unexpected results may be produced, please pass a numeric vector.

Setting direction: controls < cases
```

```
plot(log_roc_curve.lasso)
```




```
auc(log_roc_curve.lasso)
```

```
Area under the curve: 0.6281
```

Ridge

```
model.quit.ridge <- cv.glmnet(X_train_processed, yq_train, family = binomial, alpha = 0, lam
```

```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
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Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```
model.quit.ridge_coef <- predict(model.quit.ridge, s = model.quit.ridge$lambda.min, type = "

model.quit.ridge_coefs <- as.matrix(model.quit.ridge_coef)
model.quit.ridge_vars <- data.frame(
  Variable = rownames(model.quit.ridge_coefs),
  Coefficient = as.numeric(model.quit.ridge_coefs)
) %>% filter(Coefficient != 0 & Variable != "(Intercept)") %>%
  arrange(Coefficient)


model.quit.ridge_vars
```

```
                                  Variable   Coefficient
1               _advice_to_management_focus -0.1102993986
2                         _review_cons_back -0.1065360082
3                        _review_cons_covid -0.1056774131
4                    _review_cons_regarding -0.1033512508
5            _advice_to_management_employee -0.0996149438
6                     _review_pros_software -0.0984593318
7               _advice_to_management_allow -0.0929003242
8               _advice_to_management_great -0.0880746604
9                        _review_pros_goals -0.0854541706
10                        _review_pros_work -0.0830307198
```

```
11                 _review_cons_companies -0.0783518163
12                 _review_cons_company's -0.0781497257
13           _advice_to_management_senior -0.0768631736
14           _advice_to_management_people -0.0756384958
15                   _review_pros_average -0.0746129227
16                      _review_pros_okay -0.0741619786
17             _advice_to_management_high -0.0717592578
18                 _review_cons_diversity -0.0712884721
19               _review_pros_advancement -0.0705348617
20                   _review_cons_systems -0.0700636151
21            _advice_to_management_leave -0.0697147123
22             _advice_to_management_take -0.0696150637
23                    _review_pros_salary -0.0680340601
24                      _review_pros_name -0.0672492061
25                 _review_cons_processes -0.0668933338
26                 _review_cons_positions -0.0653904969
27                _review_pros_decent_pay -0.0652018619
28                     _review_cons_often -0.0649795948
29                     _review_pros_great -0.0632369776
30              _advice_to_management_job -0.0625023517
31                    _review_pros_plenty -0.0620562400
32          _advice_to_management_company -0.0618559565
33                  _review_cons_requires -0.0618404470
34                         _review_cons_3 -0.0618352632
35                     _review_pros_right -0.0615884338
36                         _summary_covid -0.0612563466
37       _advice_to_management_understand -0.0608624974
38                   _review_pros_options -0.0607774112
39                     _review_pros_brand -0.0606837862
40                     _review_cons_level -0.0605543596
41                      _review_cons_done -0.0602341158
42          _advice_to_management_workers -0.0601468317
43                _review_pros_leadership -0.0596936967
44              _review_pros_professional -0.0593917316
45                     _review_cons_sales -0.0581796178
46                    _review_pros_senior -0.0578289239
47                _review_cons_operations -0.0577375551
48             _advice_to_management_line -0.0574459802
49                   _review_cons_culture -0.0569055619
50           _advice_to_management_issues -0.0567699350
51                     _review_cons_still -0.0564211308
52             _advice_to_management_work -0.0564193264
53                     _review_pros_covid -0.0560124977
```

```
                        _review_cons_2 -0.0556861521
                   _review_pros_career -0.0555213047
                    _review_pros_lunch -0.0542127097
        _advice_to_management_processes -0.0541528265
                      _summary_culture -0.0539080402
          _advice_to_management_without -0.0539013750
              _review_cons_expectations -0.0538885248
                      _review_cons_well -0.0537552931
           _advice_to_management_career -0.0533242638
                  _review_pros_support -0.0532362683
                       _review_pros_hr -0.0530427884
               _review_cons_development -0.0528978573
                      _review_pros_take -0.0528537140
                      _review_cons_real -0.0527665893
                       _summary_school -0.0527405751
             _advice_to_management_time -0.0525589671
             _advice_to_management_hard -0.0521596022
                      _review_pros_even -0.0517619514
           _advice_to_management_ensure -0.0514675737
                      _review_cons_lead -0.0514166622
         _advice_to_management_business -0.0512061270
                      _review_cons_take -0.0511534758
                      _review_cons_year -0.0510206134
            _advice_to_management_daily -0.0510087464
                 _review_pros_genuinely -0.0509835632
          _advice_to_management_product -0.0509494487
                   _review_cons_rather -0.0506830351
                     helpfulness_score -0.0505540532
             _advice_to_management_good -0.0502453766
               _summary_company_culture -0.0498568641
                 _review_cons_different -0.0495890729
              _advice_to_management_get -0.0494782473
                      _review_cons_slow -0.0494098136
                covid_related_review_X1 -0.0489782585
                      _review_cons_days -0.0486685775
                  _review_cons_negative -0.0486559540
                 _review_cons_insurance -0.0485557753
                      _review_pros_hard -0.0485546059
       _advice_to_management_management -0.0484608190
                  _review_pros_teamwork -0.0483780917
                  _review_pros_security -0.0483252900
                        _review_cons_us -0.0482355074
                       _review_cons_two -0.0481307413
```

```
97                _review_cons_incompetent -0.0479621190
98                       _review_pros_full -0.0479010370
99                           count_helpful -0.0475926372
100                      _review_cons_used -0.0474402086
101    _advice_to_management_communication -0.0474168389
102                     _review_pros_smart -0.0469321931
103                  _review_pros_discount -0.0468321503
104        _advice_to_management_implement -0.0466066233
105                 _review_cons_resources -0.0464480913
106                       _review_pros_ceo -0.0462961934
107                      _review_pros_grow -0.0462757301
108     _advice_to_management_professional -0.0459943988
109                 _review_pros_coworkers -0.0457719324
110                     _review_cons_hours -0.0451930013
111                     _review_pros_bonus -0.0451708814
112                           _summary_old -0.0445093564
113                    _review_pros_offers -0.0442419494
114               _review_cons_competitive -0.0441544147
115                      _review_cons_code -0.0441001204
116                  _review_pros_work_home -0.0440617791
117            _advice_to_management_level -0.0439281030
118                      _review_pros_best -0.0437647079
119          _advice_to_management_culture -0.0436985334
120            _summary_great_company_work -0.0433719203
121                     _review_cons_teams -0.0432635101
122             _advice_to_management_care -0.0431153765
123                      _review_cons_hold -0.0429198081
124           _advice_to_management_growth -0.0428345838
125           _advice_to_management_talent -0.0427141610
126                      _review_pros_sick -0.0426151009
127            _advice_to_management_learn -0.0425740701
128                     _review_cons_trust -0.0422495731
129                 employee_job_start_year -0.0422062240
130                    _review_pros_pension -0.0422045493
131            _advice_to_management_upper -0.0419362124
132                    _review_cons_salary -0.0418831711
133        _review_pros_competitive_salary -0.0417603802
134                        _summary_family -0.0414270983
135                         _summary_needs -0.0414033279
136                          _summary_best -0.0413939993
137                      _review_cons_hire -0.0413680484
138                  _review_pros_different -0.0412711320
139                  _review_cons_financial -0.0411679163
```

```
                _review_cons_customers -0.0409276507
                    _review_cons_calls -0.0405683135
                    _review_cons_think -0.0404420418
                      _review_cons_see -0.0399953866
              _review_pros_flexibility -0.0399760469
                 _review_pros_projects -0.0398446543
          _advice_to_management_higher -0.0398423092
                   _review_cons_higher -0.0397684719
         _advice_to_management_clearly -0.0396919683
                       _summary_fast -0.0395962840
                     _review_pros_home -0.0394920099
                     _review_cons_make -0.0394405017
                   _review_cons_change -0.0393166191
                  _summary_worst_place -0.0392842451
                  _review_pros_offered -0.0391114920
            _review_pros_opportunities -0.0389819440
                   _review_pros_global -0.0386739360
                       _review_cons_30 -0.0385391213
                          _summary_low -0.0385179091
                      _summary_low_pay -0.0385179085
                    _review_cons_ideas -0.0385027050
             _summary_work_environment -0.0384722628
             _review_pros_pay_benefits -0.0384361791
            _advice_to_management_lack -0.0384329519
          _advice_to_management_single -0.0382115985
                    _summary_corporate -0.0379816913
                     _review_cons_work -0.0378163838
              _review_pros_competitive -0.0377932566
                _review_cons_technical -0.0377542242
                      _summary_working -0.0377252308
                   _review_pros_better -0.0377054851
                      _review_cons_run -0.0375414149
              _review_pros_good_people -0.0374534674
                        _review_cons_7 -0.0373931623
                   _review_pros_really -0.0371284021
                   _review_cons_notice -0.0369195752
                   _review_cons_issues -0.0368638767
                     _review_cons_even -0.0368310517
              _advice_to_management_hr -0.0367875903
                   _review_cons_forced -0.0367274896
                 _review_cons_personal -0.0367219584
                     _review_cons_like -0.0363915556
                      _summary_overall -0.0361806145
```

```
                  _review_cons_market -0.0360607455
                  _review_cons_growth -0.0360314405
           _advice_to_management_just -0.0359941629
            _review_pros_compensation -0.0359566050
                   _review_cons_based -0.0356657119
                        _summary_high -0.0353946770
      _advice_to_management_employees -0.0352998467
          _advice_to_management_sales -0.0352911122
                 _review_cons_results -0.0352326217
                _review_pros_pay_good -0.0348080702
               _summary_great_company -0.0346935858
                      _summary_balance -0.0345548324
           _advice_to_management_keep -0.0345155598
                    _review_cons_stay -0.0343018908
                    _review_pros_jobs -0.0342649645
                _review_pros_provides -0.0342643479
                        _summary_just -0.0339915209
                   _review_cons_means -0.0339444575
        _advice_to_management_process -0.0338898956
                  _review_cons_credit -0.0337671215
                     _review_cons_low -0.0328358408
    _advice_to_management_performance -0.0328340762
                    _review_cons_role -0.0327735016
                _review_cons_projects -0.0326105931
             _review_cons_progression -0.0324385090
                _review_cons_balance -0.0322396927
               _review_pros_including -0.0317257217
                    _review_cons_jobs -0.0316550153
                  _review_pros_direct -0.0316283739
             _review_pros_progression -0.0316102356
                    _review_cons_full -0.0314067173
                   _review_cons_night -0.0311487346
                      _summary_senior -0.0309597516
                        _summary_poor -0.0308799946
          _review_cons_unprofessional -0.0307245621
                     _review_pros_day -0.0307060792
                    _review_pros_year -0.0305763850
                    _review_cons_life -0.0304994080
                _review_cons_benefit -0.0304031514
               _review_pros_fantastic -0.0302570633
                    _review_cons_seen -0.0300988288
                     _summary_manager -0.0299815605
                _review_pros_leaders -0.0298572729
```

```
                 _review_cons_meeting -0.0296629932
              _review_pros_colleagues -0.0294319458
              _review_pros_great_work -0.0292113638
        _advice_to_management_working -0.0289237067
                 _review_pros_working -0.0287508032
            _advice_to_management_can -0.0286252257
                  _review_pros_pretty -0.0284922495
                 _review_cons_overall -0.0284375179
        _advice_to_management_address -0.0282815415
           _review_cons_opportunities -0.0280875070
              _review_cons_promotions -0.0278555462
                _review_pros_friendly -0.0277264268
              _review_pros_supportive -0.0275999315
    _review_pros_compensation_benefits -0.0275297440
                        _summary_good -0.0274408824
            _review_pros_good_company -0.0273356766
                  _review_cons_actual -0.0270899638
            _review_cons_compensation -0.0270169074
       _advice_to_management_everyone -0.0268443902
                       _review_cons_6 -0.0267798841
        _advice_to_management_instead -0.0267583850
                     _review_cons_top -0.0267311998
                  _review_cons_months -0.0267078470
               _review_pros_insurance -0.0266797690
                 _review_cons_treated -0.0266586996
                    _review_cons_give -0.0264995302
           _review_pros_benefits_good -0.0264470317
               _review_cons_sometimes -0.0264102885
              _review_cons_inadequate -0.0261980515
                      _review_pros_job -0.0257594187
                  _review_cons_proper -0.0254517808
         _advice_to_management_better -0.0254330025
                    _review_pros_plan -0.0252024313
                _summary_environment -0.0250779492
                    _review_pros_life -0.0249413979
             _review_pros_remote_work -0.0248081578
                   _review_cons_years -0.0243443008
                 _summary_experience -0.0233799047
                 _review_pros_growth -0.0231224308
                   _review_cons_week -0.0229611361
                _summary_organization -0.0229275721
                       _summary_place -0.0228231742
                     _review_pros_new -0.0227445781
```

```
          _advice_to_management_salary -0.0226954386
                     _summary_company -0.0224749724
             _advice_to_management_new -0.0221667435
             _review_pros_nice_people -0.0215773783
                      count_unhelpful -0.0210810320
                 _review_cons_process -0.0209198735
                   _review_cons_goals -0.0205965180
                _review_pros_industry -0.0205053976
               _summary_opportunities -0.0204673840
                   _review_pros_large -0.0201994824
                _summary_company_work -0.0198886217
                  rating_work_life_X4 -0.0197974073
                    _review_pros_long -0.0189041618
               _review_cons_colleagues -0.0188729200
         _review_pros_benefits_package -0.0186917581
                    _summary_terrible -0.0183956562
               _review_pros_co_workers -0.0182739413
                     _review_cons_day -0.0182617624
                   _review_pros_money -0.0181382011
                          rating_year -0.0181053724
                        _summary_work -0.0175424436
             _review_cons_life_balance -0.0172727405
        _review_cons_work_life_balance -0.0172727402
       rating_compensation_benefits_X4 -0.0170655042
                 _review_pros_business -0.0170240752
                  _review_cons_listen -0.0169201402
                    rating_overall_X5 -0.0168683605
        rating_career_opportunities_X4 -0.0166858453
             _review_pros_good_working -0.0165899560
        rating_career_opportunities_X5 -0.0162710888
                     _review_cons_due -0.0160779227
                  _review_pros_office -0.0158800100
                  _review_pros_higher -0.0157920285
                 _review_cons_without -0.0156864569
                 _review_cons_outside -0.0156817759
               _review_pros_management -0.0154939252
                _review_cons_work_life -0.0154442037
                _review_cons_problems -0.0153457210
                _review_cons_company -0.0153003808
                 _review_cons_raises -0.0151538302
          _advice_to_management_listen -0.0151066084
                     _review_cons_self -0.0147023512
                  _review_cons_office -0.0146224548
```

```
                    _review_pros_fun -0.0141118990
                _review_cons_leaving -0.0141008400
                _review_pros_workers -0.0140813654
                    _review_cons_one -0.0140606166
       rating_diversity_inclusion_X5 -0.0133935037
            _review_cons_experienced -0.0132716420
                _review_cons_layoffs -0.0130646906
                _review_pros_product -0.0124181506
            rating_culture_values_X4 -0.0123414345
                    _review_cons_e.g -0.0123389684
      flags_recommend_frend_POSITIVE -0.0121100321
                  _review_cons_heavy -0.0118149614
     flags_business_outlook_POSITIVE -0.0117863145
            rating_culture_values_X2 -0.0115278842
                     _review_pros_co -0.0114443938
                _summary_leadership -0.0111274530
              _review_cons_planning -0.0110952104
              _review_cons_feedback -0.0108612523
                _review_pros_remote -0.0108480188
              _review_pros_good_pay -0.0106696653
                _review_cons_profit -0.0100466127
    _review_pros_working_environment -0.0099640648
                 _summary_stay_away -0.0098742720
              _review_pros_creative -0.0096599167
                 _review_cons_leave -0.0095274816
               _review_cons_changes -0.0089558092
                rating_work_life_X5 -0.0088206847
                 _review_cons_lower -0.0085613377
            _summary_poor_management -0.0082152400
             _review_cons_management -0.0082135371
                 _review_cons_great -0.0080900040
                  rating_overall_X3 -0.0079684542
              _review_cons_customer -0.0075719099
         rating_senior_leadership_X4 -0.0075132948
              _review_pros_training -0.0074881834
                  _review_pros_time -0.0071231386
             _review_cons_political -0.0066993796
                 _summary_employees -0.0062003844
                  _review_cons_time -0.0061035039
                 _review_cons_roles -0.0058349417
                  _review_cons_high -0.0056534311
              _summary_good_company -0.0055892232
                 _review_cons_times -0.0051935075
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                 _review_cons_service -0.0050028757
        rating_diversity_inclusion_X3 -0.0048163762
             rating_culture_values_X5 -0.0047531192
                 _review_cons_limited -0.0046308496
                        _summary_stay -0.0046303456
            _summary_great_place_work -0.0038578858
                    _review_cons_lack -0.0037895241
                     _review_pros_pay -0.0037432234
                 _summary_great_place -0.0036385650
        rating_diversity_inclusion_X2 -0.0035578172
_advice_to_management_opportunities -0.0034070886
                   _review_cons_tools -0.0033708915
                  _review_pros_though -0.0032520567
                _review_pros_learning -0.0027158007
                       _review_cons_5 -0.0026360449
               _review_pros_customers -0.0022276085
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                    rating_overall_X2 -0.0020465649
      rating_compensation_benefits_X3 -0.0019918098
          rating_senior_leadership_X3 -0.0018538552
      rating_compensation_benefits_X2 -0.0016717244
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        _review_pros_environment_good -0.0006887036
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                    _review_cons_left -0.0003748218
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        rating_diversity_inclusion_X4  0.0001336033
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              _review_pros_technology  0.0020893634
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                       _summary_learn  0.0024738761
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           _advice_to_management_stop  0.0056549748
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                    _review_cons_less  0.0061635676
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                _review_cons_support  0.0073869927
               _review_pros_providing  0.0076498186
                _review_cons_pressure  0.0077817941
                _review_cons_nothing  0.0079193321
                    _review_cons_good  0.0085720295
               _review_cons_extremely  0.0085868402
                _review_cons_regional  0.0086813669
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                    _review_pros_know  0.0095796438
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                _review_cons_working  0.0099522157
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                    _review_cons_told  0.0109055804
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           _advice_to_management_many  0.0120488590
                    _review_pros_free  0.0125747459
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                   _review_cons_among  0.0137060518
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441          _review_cons_micromanagement  0.0137451851
442                _review_cons_excessive  0.0139266583
443                 _review_cons_terrible  0.0139550618
444         _advice_to_management_leaders  0.0159431069
445                       _review_cons_let  0.0163468537
446                          _summary_team  0.0166355061
447                       _review_pros_may  0.0167380092
448                _review_cons_difficult  0.0170536300
449                    _review_pros_level  0.0171012107
450                     _review_cons_need  0.0174670974
451                  _review_cons_instead  0.0176721208
452                   _review_cons_little  0.0177921154
453                    _review_cons_upper  0.0181777164
454                   _review_pros_around  0.0183962625
455               _review_cons_favoritism  0.0186056321
456                          _summary_okay  0.0188868672
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458                    _review_cons_money  0.0195220384
459              _review_cons_individuals  0.0199236236
460                         _summary_worst  0.0202314407
461                  _review_cons_provided  0.0204755192
462                    _review_cons_period  0.0207888082
463                   _review_pros_package  0.0208798956
464                        _review_cons_hr  0.0209439411
465         _advice_to_management_provide  0.0211698069
466            _advice_to_management_need  0.0211806419
467          _advice_to_management_reduce  0.0213740554
468                       _review_cons_20  0.0217856425
469                _review_pros_employees  0.0221316667
470                  _review_cons_internal  0.0223552927
471               _review_cons_environment  0.0230200284
472                  _review_cons_horrible  0.0237474919
473                      _review_pros_care  0.0240779838
474         flags_ceo_approval_DISAPPROVE  0.0243849779
475           _review_pros_good_work_life  0.0244483417
476                       _summary_growth  0.0244968118
477                   _review_cons_average  0.0246346226
478                     _review_cons_cares  0.0247692147
479                _review_pros_work_life  0.0247930254
480                      _review_pros_site  0.0257933324
481                     _review_pros_learn  0.0258382346
482               _review_pros_opportunity  0.0262308202
483             _review_cons_communication  0.0262499188
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                         _summary_bad  0.0277856830
          _advice_to_management_staff  0.0285264263
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                _review_cons_business  0.0293449526
                  _review_cons_values  0.0294038770
                   _review_cons_lacks  0.0298143944
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       _review_cons_toxic_environment  0.0309876505
                    _review_cons_made  0.0312010475
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           _review_pros_good_benefits  0.0315172666
                 career_missing_TRUE.  0.0318492014
                  _summary_place_work  0.0320391848
          _advice_to_management_treat  0.0325632713
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                _review_cons_meetings  0.0331237500
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                _review_cons_multiple  0.0346522897
                   _review_pros_place  0.0350706959
                    _review_cons_term  0.0358389416
                   _review_cons_short  0.0361377209
       _review_cons_senior_management  0.0363776757
                    _review_pros_tech  0.0373307187
                     _review_cons_per  0.0377615425
                  _review_pros_people  0.0383279769
        _advice_to_management_respect  0.0389118464
                    _review_pros_many  0.0392422082
                  _review_pros_things  0.0394782636
                _review_cons_politics  0.0394871942
              _review_cons_everything  0.0398096107
                   _review_cons_focus  0.0399021159
                     _review_pros_lot  0.0399826130
                  _review_cons_manager  0.0399933994
                _review_cons_outdated  0.0400175379
                  _review_cons_reviews  0.0405628680
                _review_cons_products  0.0413691224
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               _review_cons_corporate  0.0434168665
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                  _review_pros_decent  0.0451317471
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                _review_cons_behavior  0.0480579496
               _review_pros_good_work  0.0485752506
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                     _review_cons_go  0.0487459138
            _review_pros_good_salary  0.0487944314
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      rating_diversity_inclusion_X4.5  0.0525006845
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574               _review_cons_department  0.0542308721
575                      _summary_lacking  0.0542698699
576                        _summary_money  0.0544142670
577                     _review_cons_shift  0.0545506682
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579                        _summary_awful  0.0547147867
580                        _summary_think  0.0554034349
581                    _summary_workplace  0.0557308632
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583                         _summary_look  0.0562761459
584                        _summary_avoid  0.0565563722
585                         _summary_time  0.0568652470
586                _review_pros_generally  0.0572888547
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588                     _review_cons_right  0.0577898096
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590                  _summary_place_start  0.0582458976
591                     _review_pros_often  0.0584479683
592                _review_cons_decisions  0.0585135302
593                     _review_cons_toxic  0.0585491454
594                  _review_cons_product  0.0587238230
595         _review_pros_good_pay_benefits  0.0590871005
596                       _review_cons_job  0.0591406947
597                       _review_cons_ceo  0.0591673725
598                      _review_cons_poor  0.0592239424
599                      _review_pros_like  0.0606029810
600              _summary_worst_place_work  0.0607047710
601                        _summary_review  0.0607055211
602           _review_pros_good_place_work  0.0608594575
603              _advice_to_management_pay  0.0610844655
604                  _review_cons_reviewer  0.0611094564
605            _advice_to_management_teams  0.0611437617
606                    _review_cons_senior  0.0612029211
607                  _review_cons_program  0.0613240347
608                _review_pros_technical  0.0613529476
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612           _review_pros_decent_benefits  0.0616749789
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_review_cons_owners	0.0678020850
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_advice_to_management_individuals	0.0750996948

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                _review_cons_reality  0.0806880915
                   _review_pros_kind  0.0809715769
              _review_cons_marketing  0.0810746221
                  _review_pros_meet  0.0811507257
              _review_pros_available  0.0812224294
             _review_pros_healthcare  0.0812358884
                  _review_pros_rate  0.0812851849
                 _review_pros_still  0.0813043163
                _review_pros_family  0.0814525461
               _review_pros_offering  0.0817576158
               _review_pros_holidays  0.0817763851
                _review_pros_helpful  0.0819243792
                   _review_pros_paid  0.0821788487
                _review_pros_medical  0.0824904423
                 _review_pros_travel  0.0828935405
         _review_cons_poor_management  0.0832114459
               _review_cons_properly  0.0834524149
                     _summary_unless  0.0834600153
             _review_pros_commission  0.0835259107
                _review_cons_bonuses  0.0836525256
        _review_cons_upper_management  0.0836713865
            _advice_to_management_due  0.0837091318
             _review_pros_especially  0.0837456298
```

_review_cons_departments	0.0837714490
_review_cons_long	0.0839537166
_summary_employee	0.0840244487
_review_pros_opportunity_learn	0.0840345686
_summary_career	0.0841162538
_review_pros_benefits	0.0845083046
_review_pros_rewarding	0.0846926129
_review_cons_training	0.0849188725
_review_cons_ever	0.0849504449
_review_cons_expect	0.0849900697
_review_pros_unlimited	0.0850200277
_review_pros_staff	0.0850488172
_review_cons_breaks	0.0850731075
_review_cons_avoid	0.0850866078
_review_cons_things	0.0854938002
_advice_to_management_numbers	0.0855045279
_review_cons_promotion	0.0857769697
_review_cons_first	0.0858495821
_review_cons_revenue	0.0858752950
_review_pros_enjoyed	0.0860264954
_review_pros_schedule	0.0860426313
_review_cons_long_term	0.0861521277
_review_cons_required	0.0866421442
_review_cons_gossip	0.0867384544
_review_pros_overtime	0.0872488325
_review_cons_makes	0.0875156485
_review_pros_wonderful	0.0879854681
_review_cons_getting	0.0881326855
_review_cons_benefits	0.0882302287
_review_cons_tech	0.0883732393
_review_pros_caring	0.0884797106
_review_pros_provided	0.0885581340
_review_cons_month	0.0885739456
_review_cons_payment	0.0887506104
_review_pros_impact	0.0887937258
_review_cons_family	0.0890785852
_review_cons_quickly	0.0891828737
_review_cons_promoted	0.0892113735
_review_cons_position	0.0895184058
_review_cons_promised	0.0895357950
_advice_to_management_needs	0.0895371267
_review_pros_cool	0.0896705738
_review_pros_events	0.0900234957

_review_cons_4	0.0902355137
_review_cons_mental	0.0903559112
_review_cons_though	0.0903849556
_review_pros_make	0.0904609679
_review_pros_allowing	0.0905186375
_review_pros_products	0.0905206197
_review_cons_overtime	0.0907223660
_review_cons_many	0.0912630443
_review_pros_great_people	0.0914016349
_review_cons_members	0.0916136747
_review_cons_system	0.0916404393
_review_cons_advancement	0.0919244062
_review_cons_given	0.0920567309
_review_cons_small	0.0920995463
_review_pros_mission	0.0921663027
_review_cons_always	0.0922429171
_summary_ever	0.0922927680
_review_pros_learning_opportunities	0.0924435194
_review_cons_last	0.0924788275
_review_cons_rude	0.0929362031
_review_cons_employee	0.0929390359
_advice_to_management_based	0.0929668112
_review_pros_401k	0.0932159199
_review_pros_friends	0.0933197534
_review_cons_school	0.0933626825
_review_cons_stress	0.0934058582
_review_pros_talented	0.0935585718
_review_cons_lot	0.0937560995
_review_cons_frequent	0.0938968200
_review_pros_tools	0.0942767141
_review_pros_culture	0.0943038904
_review_pros_amazing	0.0945051302
_review_pros_environment	0.0945491634
_review_cons_numbers	0.0946287654
_review_cons_new	0.0947573250
_review_cons_just	0.0948135465
_review_pros_atmosphere	0.0950528245
_review_cons_underpaid	0.0951898417
_review_pros_positive	0.0952294750
_review_pros_employee	0.0952898196
_review_pros_worked	0.0960426153
_review_cons_hiring	0.0961955442
_review_cons_almost	0.0961958318

_review_pros_interesting	0.0964138063
_summary_start	0.0964932897
_review_pros_great_benefits	0.0969226524
_review_pros_vacation	0.0969336781
_review_cons_quality	0.0970653808
_review_pros_areas	0.0972322652
_review_cons_everyone	0.0973622616
_review_cons_rules	0.0980711395
_review_cons_takes	0.0985030542
_review_pros_skills	0.0985034307
_review_cons_turnover	0.0987193852
_review_cons_unless	0.0988365507
_review_cons_job_security	0.0994300669
_review_cons_way	0.0994474396
_review_cons_overworked	0.0995226130
_review_cons_room	0.0999424649
_review_cons_also	0.1001775818
_review_cons_holidays	0.1008333291
_review_cons_accounts	0.1008778462
_review_cons_sick	0.1013121182
_review_cons_hard	0.1014458649
_review_cons_promises	0.1018979683
_review_pros_people_work	0.1019635235
_review_cons_workers	0.1020614818
_review_cons_care_employees	0.1022716388
_review_cons_may	0.1022753143
_review_cons_complete	0.1023686903
_review_cons_organization	0.1030973035
_advice_to_management_give	0.1037699302
_review_cons_laid	0.1038781647
_review_cons_weeks	0.1038922054
_review_cons_never	0.1043123831
_review_pros_development	0.1045787919
_review_cons_fired	0.1064411031
_review_cons_value	0.1090541317
_review_cons_security	0.1096711943
_review_cons_every	0.1096896507
_review_cons_clients	0.1107634658
_review_cons_daily	0.1118701093
_review_pros_one	0.1168113488
_review_cons_place	0.1168555786
_review_cons_others	0.1180557276
_review_pros_hours	0.1198891216

```
957                      _review_cons_zero  0.1207202788
958                  _review_cons_managers  0.1238472827
```

```r
roc_obj <- roc(yq_train, predict(model.quit.ridge, newx = X_train_processed, type = "respons
```

```
Setting levels: control = 0, case = 1
```

```
Warning in roc.default(yq_train, predict(model.quit.ridge, newx =
X_train_processed, : Deprecated use a matrix as predictor. Unexpected results
may be produced, please pass a numeric vector.
```

```
Setting direction: controls < cases
```

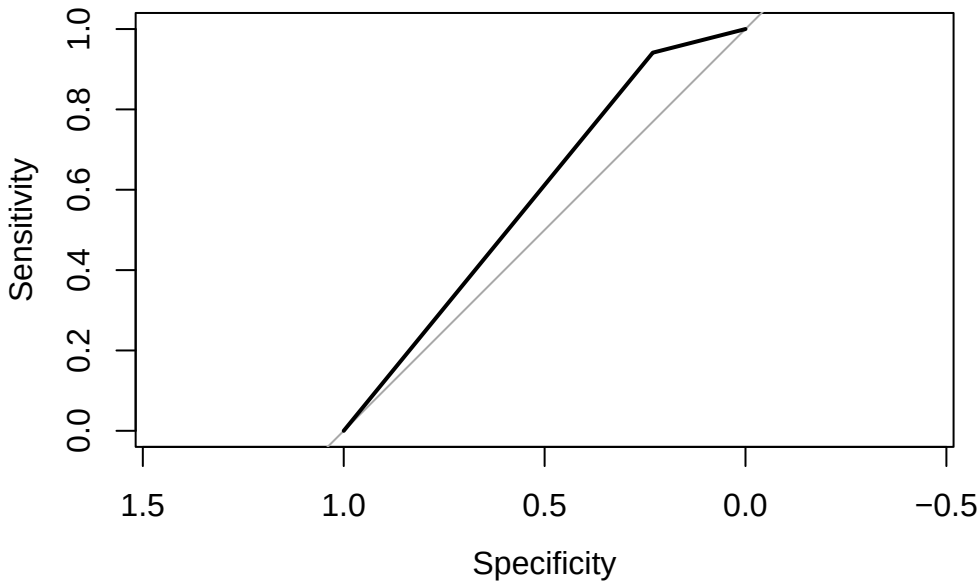
```r
best_threshold.ridge <- coords(roc_obj, "best", ret = "threshold")[1,1]

yq_pred.ridge <- ifelse(predict(model.quit.ridge, newx = X_test_processed, type = "response")
log_roc_curve.ridge <- roc(yq_test, yq_pred.ridge)
```

```
Setting levels: control = 0, case = 1
```

```
Warning in roc.default(yq_test, yq_pred.ridge): Deprecated use a matrix as
predictor. Unexpected results may be produced, please pass a numeric vector.
```

```
Setting direction: controls < cases
```

```r
plot(log_roc_curve.ridge)
```




```
auc(log_roc_curve.ridge)
```

```
Area under the curve: 0.586
```

Elastic Net

```
results = list()
for(a in seq(0,1,0.05)){
  model.quit.elastic <- cv.glmnet(X_train_processed, yq_train, family = binomial, alpha = a,
  results[[as.character(a)]] <- min(model.quit.elastic$cvm)
}
```

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Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```r
# find best alpha
best_alpha <- names(which.min(unlist(results)))
print(paste("The best Alpha is:", best_alpha))
```

```
[1] "The best Alpha is: 0.8"
```

```r
model.quit.elastic <- cv.glmnet(X_train_processed, yq_train, family = binomial, alpha = best_

model.quit.elastic_coef <- predict(model.quit.elastic, s = model.quit.elastic$lambda.min, typ

model.quit.elastic_coefs <- as.matrix(model.quit.elastic_coef)
model.quit.elastic_vars <- data.frame(
  Variable = rownames(model.quit.elastic_coefs),
  Coefficient = as.numeric(model.quit.elastic_coefs)
) %>% filter(Coefficient != 0 & Variable != "(Intercept)") %>%
```

```
  arrange(Coefficient)


model.quit.elastic_vars
```

```
                         Variable  Coefficient
1              _review_pros_covid -1.790353368
2              _review_cons_covid -1.474711075
3               _review_pros_name -1.437811527
4         covid_related_review_X1 -0.914125729
5              _review_cons_still -0.293124400
6    _advice_to_management_ensure -0.033979811
7 _advice_to_management_processes -0.004479564
8            career_missing_TRUE.  0.328456874
```

```
roc_obj <- roc(yq_train, predict(model.quit.elastic, newx = X_train_processed, type = "respon
```

```
Setting levels: control = 0, case = 1
```

```
Warning in roc.default(yq_train, predict(model.quit.elastic, newx =
X_train_processed, : Deprecated use a matrix as predictor. Unexpected results
may be produced, please pass a numeric vector.
```

```
Setting direction: controls < cases
```

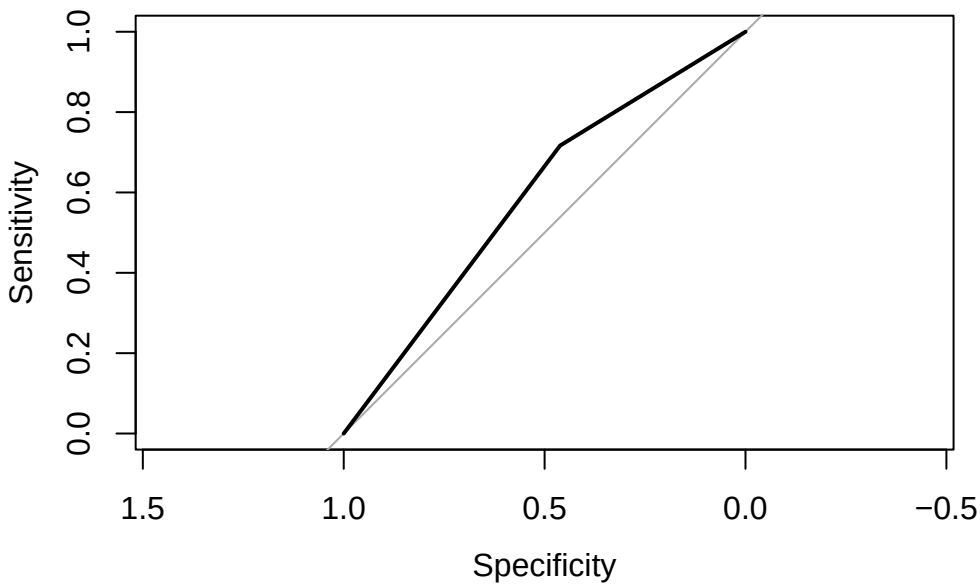
```
best_threshold.elastic <- coords(roc_obj, "best", ret = "threshold")[1,1]

yq_pred.elastic <- ifelse(predict(model.quit.elastic, newx = X_test_processed, type = "respon
log_roc_curve.elastic <- roc(yq_test, yq_pred.elastic)
```

```
Setting levels: control = 0, case = 1
```

```
Warning in roc.default(yq_test, yq_pred.elastic): Deprecated use a matrix as
predictor. Unexpected results may be produced, please pass a numeric vector.
```

```
Setting direction: controls < cases
```

```r
plot(log_roc_curve.elastic)
```




```r
auc(log_roc_curve.elastic)
```

```
Area under the curve: 0.5891
```

Report

```r
get_metrics_bin <- function(y_true, y_pred, name,
    positive = "1") {
  cm_obj <- confusionMatrix(
    data = as.factor(y_pred),
    reference = as.factor(y_true),
    positive = positive
  )
  out <- c(
    Sensitivity = unname(cm_obj$byClass["Sensitivity"]),
    Specificity = unname(cm_obj$byClass["Specificity"]),
    Accuracy    = unname(cm_obj$overall["Accuracy"]),
    AOC = auc(roc(yq_test, yq_pred.ridge))
  )
  return (c(Name = name, round(out, 4)))
```

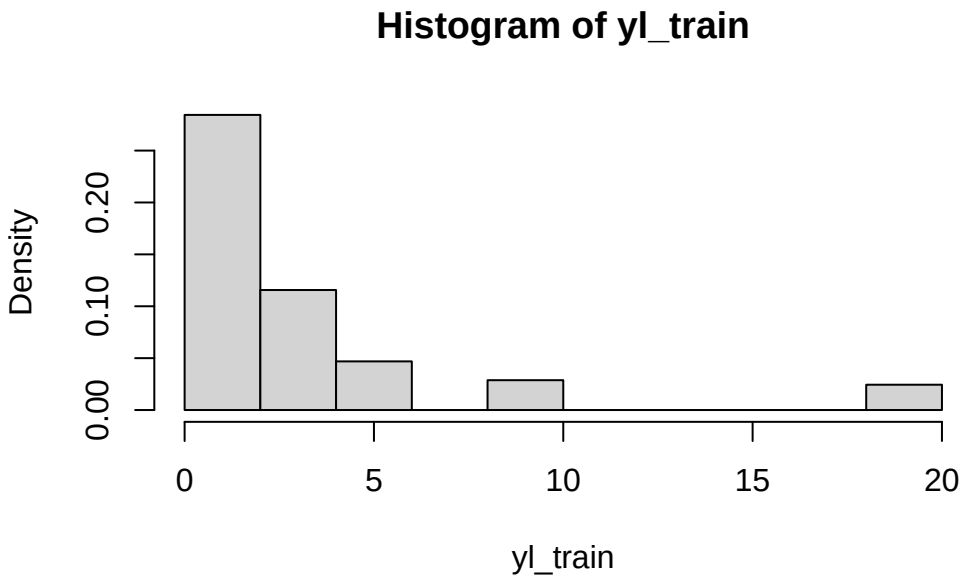
```r
}


rbind(
  get_metrics_bin(yq_test, yq_pred.lasso, "Lasso-Logistics"),
  get_metrics_bin(yq_test, yq_pred.ridge, "Ridge-Logistics"),
  get_metrics_bin(yq_test, yq_pred.elastic, "Elastic-Logistics")

)
```

```
Setting levels: control = 0, case = 1

Warning in roc.default(yq_test, yq_pred.ridge): Deprecated use a matrix as
predictor. Unexpected results may be produced, please pass a numeric vector.

Setting direction: controls < cases
Setting levels: control = 0, case = 1

Warning in roc.default(yq_test, yq_pred.ridge): Deprecated use a matrix as
predictor. Unexpected results may be produced, please pass a numeric vector.

Setting direction: controls < cases
Setting levels: control = 0, case = 1

Warning in roc.default(yq_test, yq_pred.ridge): Deprecated use a matrix as
predictor. Unexpected results may be produced, please pass a numeric vector.

Setting direction: controls < cases

     Name                Sensitivity Specificity Accuracy AOC
[1,] "Lasso-Logistics"   "0.8717"    "0.3846"    "0.84"   "0.586"
[2,] "Ridge-Logistics"   "0.9412"    "0.2308"    "0.895"  "0.586"
[3,] "Elastic-Logistics" "0.7166"    "0.4615"    "0.7"    "0.586"
```

Length Model

Employee length is a fat tail variable and extremely right-skewed. Box-Cox trsnaformation applied here to deal with this problem.

```r
hist(yl_train, freq = F)
```




```r
p_load(forecast)

lambda <- BoxCox.lambda(yl_train)

print(paste("Optimal Lambda:", lambda))
```

```
[1] "Optimal Lambda: -0.37607303888529"
```

```r
yl_train2 <- BoxCox(yl_train, lambda)
yl_test2 <- BoxCox(yl_test, lambda)


hist(yl_train2, freq = F)
```

Lasso

```
lambda <- 10^seq(1,-2,length = 100)
model.length.lasso <- cv.glmnet(X_train_processed, yl_train, alpha = 1, lambda = lambda)

model.length.lasso_coef <- predict(model.length.lasso, s = model.length.lasso$lambda.min, typ

model.length.lasso_coefs <- as.matrix(model.length.lasso_coef)
model.length.lasso_vars <- data.frame(
  Variable = rownames(model.length.lasso_coefs),
  Coefficient = as.numeric(model.length.lasso_coefs)
) %>% filter(Coefficient != 0 & Variable != "(Intercept)") %>%
  arrange(Coefficient)


model.length.lasso_vars
```

```
                        Variable  Coefficient
1         employee_job_start_year -27.00650169
2      _review_pros_international  -1.34918131
3           _review_pros_managers  -0.74480841
4          _review_pros_providing  -0.49960980
5               _review_pros_nice  -0.36442980
6   _advice_to_management_support  -0.35268157
```

```
7              _review_cons_avoid  -0.32739682
8    _review_pros_good_place_work  -0.31354472
9             _review_cons_stress  -0.30785213
10          _review_pros_students  -0.28451788
11              _review_cons_many  -0.21959636
12            _review_cons_always  -0.10415143
13            rating_work_life_X3  -0.09948927
14    rating_senior_leadership_X3  -0.07652609
15             _review_cons_since   0.04663331
16 _advice_to_management_internal   0.04947896
17              _review_pros_good   0.05946340
18              _review_pros_grow   0.06598482
19            _review_cons_unless   0.11559432
20               _review_cons_two   0.28129892
21      _review_pros_life_balance   0.33380180
22          _review_cons_products   0.54184464
23              _review_cons_well   0.58152827
24        covid_related_review_X1   0.68456763
25                     rating_year   9.43830620
```

Ridge

```
model.length.ridge <- cv.glmnet(X_train_processed, yq_train, family = binomial, alpha = 0, la

model.length.ridge_coef <- predict(model.length.ridge, s = model.length.ridge$lambda.min, typ

model.length.ridge_coefs <- as.matrix(model.length.ridge_coef)
model.length.ridge_vars <- data.frame(
  Variable = rownames(model.length.ridge_coefs),
  Coefficient = as.numeric(model.length.ridge_coefs)
) %>% filter(Coefficient != 0 & Variable != "(Intercept)") %>%
  arrange(Coefficient)


model.length.ridge_vars
```

```
                                Variable   Coefficient
1                       _review_pros_name -7.596082e-01
2                       _review_cons_take -7.476293e-01
3                       _review_cons_lead -7.107042e-01
```

```
4                    _review_pros_teamwork -6.966532e-01
5          _advice_to_management_processes -6.963761e-01
6                      _review_pros_covid -6.770059e-01
7                      _review_cons_covid -6.744689e-01
8             _advice_to_management_ensure -6.410877e-01
9                 _review_cons_incompetent -6.015350e-01
10             _advice_to_management_time -5.815540e-01
11                     _review_pros_brand -5.783760e-01
12                     _review_cons_still -5.731402e-01
13                    _review_pros_senior -5.480525e-01
14                     _review_cons_level -5.048454e-01
15                      _review_cons_well -4.993778e-01
16                   _review_pros_pension -4.920722e-01
17                  _review_pros_software -4.634726e-01
18                   _review_cons_culture -4.486002e-01
19                         _review_cons_2 -4.448025e-01
20         _advice_to_management_business -4.340473e-01
21                     _review_pros_smart -4.317558e-01
22            _advice_to_management_daily -4.186781e-01
23                      _review_cons_days -4.152345e-01
24                     _review_pros_right -4.039950e-01
25                      _review_pros_take -4.028971e-01
26                      _review_pros_best -3.946410e-01
27                          _summary_fast -3.927079e-01
28           _advice_to_management_without -3.820215e-01
29                   _review_pros_support -3.747394e-01
30        _review_pros_competitive_salary -3.719754e-01
31                     _review_pros_goals -3.643939e-01
32            _advice_to_management_focus -3.631601e-01
33                _summary_company_culture -3.499714e-01
34                _review_pros_advancement -3.387207e-01
35                       _review_pros_grow -3.290514e-01
36             _advice_to_management_hard -3.264396e-01
37              _summary_work_environment -3.260418e-01
38                    _review_cons_change -3.257834e-01
39                   _summary_worst_place -3.257662e-01
40         _advice_to_management_employee -3.250116e-01
41                  _review_cons_requires -3.243407e-01
42            _advice_to_management_issues -3.199936e-01
43                _review_pros_flexibility -3.117222e-01
44                       _review_pros_work -3.072124e-01
45                        _review_cons_see -3.071575e-01
46                  _review_cons_financial -3.053693e-01
```

```
47          _advice_to_management_senior -3.039230e-01
48                      _summary_family -3.026714e-01
49         _advice_to_management_clearly -3.010938e-01
50                 _review_cons_systems -2.954872e-01
51                      _review_cons_us -2.895092e-01
52                    _review_cons_back -2.861642e-01
53           _advice_to_management_leave -2.835277e-01
54                   _review_cons_think -2.832182e-01
55               _review_cons_processes -2.807274e-01
56          _advice_to_management_career -2.795300e-01
57               _review_cons_company's -2.780887e-01
58          _advice_to_management_people -2.768235e-01
59           _summary_great_company_work -2.726482e-01
60                     _summary_overall -2.714124e-01
61                    _review_pros_full -2.707530e-01
62            _advice_to_management_lack -2.707415e-01
63                     _review_cons_run -2.654600e-01
64           _advice_to_management_allow -2.580007e-01
65                   _review_cons_trust -2.537230e-01
66                 _review_pros_average -2.472238e-01
67                     _review_pros_ceo -2.463176e-01
68                  _review_cons_salary -2.461413e-01
69                       _review_cons_3 -2.452401e-01
70                    _review_cons_slow -2.443810e-01
71               _review_cons_companies -2.440763e-01
72                    _review_cons_used -2.433069e-01
73                    _review_cons_full -2.405936e-01
74                     _summary_culture -2.374432e-01
75            _advice_to_management_just -2.313751e-01
76           _advice_to_management_great -2.294407e-01
77           _advice_to_management_learn -2.288597e-01
78               _review_cons_insurance -2.279953e-01
79       _advice_to_management_management -2.253518e-01
80                     _summary_balance -2.234247e-01
81                       _summary_covid -2.234002e-01
82               _review_pros_decent_pay -2.228713e-01
83                      _review_cons_30 -2.226923e-01
84       _advice_to_management_understand -2.210708e-01
85           _advice_to_management_growth -2.210072e-01
86                   _review_cons_means -2.170855e-01
87                    _review_cons_done -2.148930e-01
88                _review_pros_provides -2.143530e-01
89                    _review_cons_hold -2.133741e-01
```

```
                        _review_pros_hr -2.130037e-01
                 _review_pros_including -2.102912e-01
                 _review_pros_coworkers -2.100565e-01
                    _review_cons_higher -2.074943e-01
                    _review_cons_credit -2.071897e-01
                  _review_cons_projects -2.039712e-01
         _advice_to_management_implement -2.035373e-01
                       _review_pros_even -2.015279e-01
                         _summary_school -1.984252e-01
                  _review_cons_personal -1.975655e-01
                     _review_cons_based -1.934907e-01
           _advice_to_management_single -1.925545e-01
              _review_pros_competitive -1.921337e-01
    _advice_to_management_communication -1.898859e-01
                      _review_cons_stay -1.894196e-01
                     _review_cons_often -1.876406e-01
                          _summary_just -1.864483e-01
            _advice_to_management_care -1.852939e-01
                 _review_pros_friendly -1.850375e-01
                     _review_pros_great -1.838082e-01
                     _review_pros_lunch -1.818491e-01
                          _summary_high -1.809743e-01
                    _review_pros_plenty -1.809477e-01
                 _review_pros_security -1.778147e-01
             _review_pros_professional -1.774679e-01
               _review_pros_leadership -1.751467e-01
                     _review_cons_sales -1.731196e-01
                _review_cons_resources -1.700601e-01
            _advice_to_management_work -1.675531e-01
           _advice_to_management_talent -1.668662e-01
                       _review_pros_okay -1.663740e-01
                   _review_pros_offered -1.642741e-01
                _review_cons_regarding -1.639265e-01
              _review_cons_progression -1.634619e-01
                       _review_cons_low -1.617489e-01
             _review_cons_expectations -1.615752e-01
                       _review_cons_hire -1.588096e-01
                _review_pros_great_work -1.584880e-01
             _review_pros_pay_benefits -1.580776e-01
     _advice_to_management_professional -1.578628e-01
                 _review_pros_discount -1.578100e-01
             _advice_to_management_job -1.576975e-01
                       _summary_working -1.568708e-01
```

```
                    _summary_manager -1.559140e-01
               _review_cons_different -1.552729e-01
                 _review_pros_leaders -1.550871e-01
                     _review_pros_day -1.548863e-01
           _advice_to_management_high -1.530924e-01
           _advice_to_management_line -1.513823e-01
                         _summary_old -1.497921e-01
                  _review_pros_better -1.489647e-01
                 _review_pros_options -1.485000e-01
               _review_cons_customers -1.467251e-01
          _advice_to_management_sales -1.463934e-01
                  _review_pros_salary -1.456946e-01
                  _review_cons_growth -1.433325e-01
             _review_pros_progression -1.419079e-01
                  _review_pros_career -1.410832e-01
                  _review_pros_direct -1.371421e-01
      _advice_to_management_employees -1.362888e-01
                   _review_cons_teams -1.359932e-01
             _review_pros_good_people -1.336092e-01
           _advice_to_management_take -1.331394e-01
                     _summary_senior -1.329940e-01
        _advice_to_management_product -1.317933e-01
          _advice_to_management_upper -1.315590e-01
                  _review_pros_global -1.314749e-01
                    _review_pros_jobs -1.278117e-01
          _review_cons_unprofessional -1.261861e-01
                         _summary_low -1.261013e-01
                    _summary_low_pay -1.261011e-01
                   _review_pros_bonus -1.259030e-01
        _advice_to_management_workers -1.256268e-01
             _review_cons_competitive -1.254225e-01
              covid_related_review_X1 -1.233427e-01
           _advice_to_management_good -1.229797e-01
                  _review_cons_forced -1.227045e-01
                    _review_pros_hard -1.225420e-01
                       _review_cons_7 -1.220211e-01
                 _review_cons_results -1.216987e-01
               _review_pros_work_home -1.214897e-01
               _review_pros_different -1.193900e-01
                    _review_pros_sick -1.191303e-01
                   _review_cons_year -1.187721e-01
                    _review_cons_two -1.184031e-01
                  _review_cons_hours -1.180662e-01
```

```
176          _advice_to_management_better -1.171367e-01
177                     _review_cons_real -1.168118e-01
178                _review_pros_genuinely -1.164225e-01
179                     _review_cons_give -1.162532e-01
180                 _review_cons_negative -1.156798e-01
181                     helpfulness_score -1.154293e-01
182               _review_cons_operations -1.135743e-01
183                _review_cons_diversity -1.117785e-01
184                    _review_cons_ideas -1.108728e-01
185                _review_cons_positions -1.103407e-01
186                   _review_cons_issues -1.093655e-01
187                          count_helpful -1.090622e-01
188            _review_pros_benefits_good -1.082415e-01
189                     _review_pros_plan -1.080098e-01
190          _advice_to_management_higher -1.074690e-01
191                _summary_great_company -1.057155e-01
192                 _review_pros_pay_good -1.050273e-01
193                     _review_pros_home -1.050019e-01
194         _advice_to_management_address -1.042115e-01
195                _review_pros_fantastic -1.017049e-01
196                   _review_cons_rather -1.012484e-01
197           _advice_to_management_level -1.000436e-01
198                     _review_pros_life -9.968615e-02
199         _advice_to_management_company -9.878074e-02
200                          _summary_best -9.845619e-02
201         _advice_to_management_process -9.804854e-02
202            _review_pros_opportunities -9.763095e-02
203         _advice_to_management_culture -9.758578e-02
204                   _review_pros_offers -9.491309e-02
205                        _summary_needs -9.389617e-02
206               _review_cons_promotions -9.360107e-02
207      _advice_to_management_performance -9.343867e-02
208                    _review_cons_calls -9.292090e-02
209             _review_cons_compensation -9.198373e-02
210              _review_pros_remote_work -8.998593e-02
211                    _summary_corporate -8.796990e-02
212                _review_cons_technical -8.795004e-02
213                _review_pros_insurance -8.464685e-02
214                   _review_cons_notice -8.463920e-02
215                     _review_cons_jobs -8.420300e-02
216               _review_pros_supportive -8.418748e-02
217                  _summary_environment -8.305917e-02
218             _advice_to_management_get -8.270526e-02
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              _advice_to_management_hr -8.117170e-02
             _review_pros_compensation -8.100375e-02
                     _review_cons_like -8.077338e-02
               employee_job_start_year -8.015175e-02
                   _summary_experience -7.931525e-02
                         _summary_good -7.849979e-02
                _review_cons_sometimes -7.813438e-02
                          _review_cons_6 -7.619589e-02
             _review_pros_good_company -7.611453e-02
                     _review_cons_code -7.566570e-02
              _review_cons_development -7.443162e-02
                  _review_cons_balance -7.422862e-02
                     _review_cons_even -7.365512e-02
                     _review_cons_work -7.232299e-02
                   _review_pros_pretty -7.216381e-02
                  _review_pros_working -7.018865e-02
                     _review_cons_life -7.005032e-02
                   _review_pros_really -6.889442e-02
                  _review_pros_projects -6.866565e-02
                  _review_cons_overall -6.809255e-02
                     _review_pros_year -6.742865e-02
                     _review_cons_make -6.562945e-02
                   _review_pros_growth -6.469314e-02
               _review_pros_colleagues -6.436579e-02
                         _summary_poor -6.409907e-02
                _summary_opportunities -6.343207e-02
          _advice_to_management_working -6.271177e-02
                   _review_cons_months -6.202662e-02
             _advice_to_management_can -6.182522e-02
                    _review_cons_night -6.142758e-02
                   _review_cons_market -6.077712e-02
             _advice_to_management_new -5.974230e-02
                       _summary_place -5.779857e-02
                      _review_pros_job -5.711460e-02
              _review_pros_nice_people -5.674646e-02
             _review_cons_opportunities -5.673658e-02
                _review_cons_inadequate -5.602299e-02
                      _summary_company -5.562365e-02
                  rating_work_life_X4 -5.389234e-02
                     _review_cons_week -5.368461e-02
                  _review_cons_treated -5.352578e-02
         _advice_to_management_everyone -5.336130e-02
                     _review_cons_seen -5.271205e-02
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                  _review_cons_proper -5.197456e-02
                     _review_cons_top -5.063160e-02
                _review_pros_business -5.022557e-02
                   _review_cons_years -4.967080e-02
                    _review_pros_long -4.849878e-02
_review_pros_compensation_benefits -4.782150e-02
            _review_pros_co_workers -4.734979e-02
               _review_pros_industry -4.727044e-02
                     count_unhelpful -4.670072e-02
                    _review_cons_role -4.531439e-02
                _review_cons_benefit -4.509887e-02
                   _review_pros_large -4.504056e-02
                     _review_pros_new -4.501606e-02
                   rating_overall_X5 -4.469357e-02
                  _review_cons_listen -4.455773e-02
                        _summary_work -4.346446e-02
       rating_career_opportunities_X5 -4.331417e-02
                 _review_cons_process -4.268263e-02
                    _summary_terrible -4.213338e-02
      rating_compensation_benefits_X4 -4.171417e-02
            _review_pros_good_working -4.171388e-02
                _summary_company_work -4.114994e-02
         _advice_to_management_salary -4.082675e-02
       rating_career_opportunities_X4 -3.969470e-02
                 _review_cons_without -3.925627e-02
                  _review_cons_raises -3.827682e-02
                  _review_cons_actual -3.789456e-02
                _review_pros_good_pay -3.733746e-02
                     _review_cons_due -3.720155e-02
        _advice_to_management_instead -3.696369e-02
                     _review_cons_day -3.587534e-02
                          rating_year -3.586301e-02
              _review_cons_colleagues -3.533392e-02
                    _review_cons_high -3.511301e-02
                     _review_pros_fun -3.476643e-02
        _review_pros_benefits_package -3.401058e-02
                     _review_cons_one -3.311885e-02
                 _review_pros_workers -3.299501e-02
                 _review_pros_product -3.212414e-02
                      _review_pros_co -3.202424e-02
                   _summary_stay_away -3.201364e-02
                 _review_cons_outside -3.180090e-02
            _review_cons_life_balance -3.159801e-02
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305         _review_cons_work_life_balance -3.159756e-02
306               rating_culture_values_X4 -3.000584e-02
307                   _review_cons_leaving -2.997775e-02
308                   _review_cons_meeting -2.953844e-02
309                     _review_cons_goals -2.940738e-02
310                       _review_cons_e.g -2.895935e-02
311               rating_culture_values_X2 -2.832871e-02
312          rating_diversity_inclusion_X5 -2.797040e-02
313                  _review_cons_feedback -2.770847e-02
314                 _review_cons_work_life -2.714388e-02
315                    _review_pros_office -2.701841e-02
316                    _review_pros_remote -2.621175e-02
317                  _summary_organization -2.563956e-02
318                  _review_cons_problems -2.547263e-02
319                 _review_cons_management -2.466867e-02
320         flags_recommend_frend_POSITIVE -2.450851e-02
321                      _review_pros_time -2.410202e-02
322               _review_cons_experienced -2.355637e-02
323                  _review_pros_creative -2.354085e-02
324                      _review_cons_self -2.312627e-02
325        flags_business_outlook_POSITIVE -2.309656e-02
326                   _review_cons_layoffs -2.299486e-02
327                     _review_cons_leave -2.262096e-02
328                    _summary_leadership -2.251623e-02
329            _advice_to_management_listen -2.251089e-02
330                    _review_pros_higher -2.143683e-02
331                     _review_cons_heavy -2.099794e-02
332       _review_pros_working_environment -2.053097e-02
333                   _review_cons_company -1.879429e-02
334                    rating_work_life_X5 -1.851835e-02
335                     _review_pros_money -1.810829e-02
336                      rating_overall_X3 -1.780759e-02
337            rating_senior_leadership_X4 -1.743415e-02
338               _summary_poor_management -1.621865e-02
339                  _review_cons_customer -1.521734e-02
340                 _review_cons_political -1.445768e-02
341              _advice_to_management_keep -1.437839e-02
342                          _summary_stay -1.402804e-02
343          rating_diversity_inclusion_X3 -1.386858e-02
344                   _review_cons_changes -1.385782e-02
345                     _review_cons_tools -1.342000e-02
346                    _review_pros_though -1.307458e-02
347                 _review_pros_management -1.292207e-02
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                  _review_cons_service -1.192652e-02
                     _review_cons_time -1.186250e-02
                 _summary_good_company -1.128340e-02
                   _review_cons_profit -1.102780e-02
                        _review_cons_5 -1.083166e-02
                     _review_cons_talk -1.048415e-02
             _summary_great_place_work -9.779313e-03
                      _review_pros_pay -9.363230e-03
                   _review_cons_policy -8.793540e-03
       rating_compensation_benefits_X2 -8.778519e-03
                     _review_cons_lack -8.503946e-03
         rating_diversity_inclusion_X2 -7.874429e-03
                   _review_cons_common -7.139707e-03
                    _summary_employees -6.848397e-03
                     _review_cons_left -6.735235e-03
                _review_pros_customers -6.068743e-03
                _review_cons_employees -6.053113e-03
                 _review_cons_planning -5.529029e-03
                   _review_cons_office -5.270859e-03
                  _summary_great_place -5.134724e-03
                     rating_overall_X2 -4.990005e-03
             rating_culture_values_X5 -4.893161e-03
                   _review_cons_roles -4.549441e-03
                  _review_cons_limited -4.207532e-03
                 _review_pros_training -3.778099e-03
           rating_senior_leadership_X3 -3.252641e-03
               _review_pros_technology -3.164745e-03
                   _review_pros_sales -2.246726e-03
   _advice_to_management_opportunities -2.040858e-03
       rating_compensation_benefits_X3 -1.444354e-03
                 _review_cons_regional -1.292744e-03
         flags_ceo_approval_NO_OPINION -9.764772e-04
                 _review_pros_flexible -6.067854e-05
               _review_cons_low_salary  3.336119e-05
                          _summary_away  3.671136e-04
                   _review_cons_great  4.385378e-04
                    _review_pros_know  4.638475e-04
         rating_diversity_inclusion_X4  1.043179e-03
                   _review_cons_lower  1.176245e-03
                 _review_pros_learning  1.227021e-03
                _review_pros_providing  1.476195e-03
        rating_career_opportunities_X2  1.624699e-03
                     _review_cons_told  1.758777e-03
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391                _review_cons_structure  1.997683e-03
392                     _review_cons_home  2.420928e-03
393         _review_pros_environment_good  2.857934e-03
394         _advice_to_management_current  2.960803e-03
395              rating_culture_values_X3  3.715045e-03
396           rating_senior_leadership_X2  4.029969e-03
397                    _review_cons_power  4.431435e-03
398                  _review_cons_support  4.539005e-03
399                  _review_pros_seeking  4.916288e-03
400        flags_business_outlook_NEUTRAL  5.231573e-03
401                     _review_pros_can  5.812939e-03
402                _review_cons_excessive  5.821869e-03
403                    _review_cons_large  6.064421e-03
404                _review_cons_constant  6.449483e-03
405                        _summary_learn  6.467282e-03
406                     _review_pros_free  6.588871e-03
407                  _review_cons_leaders  7.593876e-03
408                     _review_pros_get  7.999662e-03
409                       _summary_great  8.122114e-03
410                       _summary_sales  8.544074e-03
411                     _review_pros_made  8.816838e-03
412                    _review_cons_times  8.852274e-03
413                     _review_cons_less  9.121728e-03
414           rating_senior_leadership_X5  9.276566e-03
415                       _summary_little  9.330531e-03
416                     _review_cons_good  1.037509e-02
417       rating_compensation_benefits_X5  1.145921e-02
418                    rating_work_life_X3  1.167855e-02
419                    _review_cons_among  1.172761e-02
420                     rating_overall_X4  1.184985e-02
421            _advice_to_management_hire  1.201354e-02
422                          _summary_team  1.230690e-02
423                    _review_cons_staff  1.236137e-02
424                _review_cons_terrible  1.335503e-02
425             _review_cons_unrealistic  1.423746e-02
426                  _review_pros_strong  1.492541e-02
427                  _review_pros_health  1.504761e-02
428                     _review_cons_let  1.546266e-02
429            _advice_to_management_like  1.582950e-02
430                  _review_cons_little  1.603426e-02
431                     _review_pros_good  1.611013e-02
432               _review_cons_extremely  1.632151e-02
433                          _summary_like  1.635182e-02
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                 _review_cons_working  1.807706e-02
              _review_cons_favoritism  1.814117e-02
                  _review_pros_annual  1.840441e-02
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                        _summary_okay  1.934173e-02
             _review_cons_environment  1.936121e-02
                  _review_cons_period  1.972359e-02
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           _advice_to_management_need  2.089385e-02
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              _review_cons_experience  2.177955e-02
         _advice_to_management_provide  2.198530e-02
             _review_cons_individuals  2.246422e-02
              _review_pros_experience  2.269774e-02
                _review_cons_instead  2.314345e-02
        _review_cons_toxic_environment  2.340085e-02
                       _summary_worst  2.381091e-02
                   _review_cons_hour  2.392809e-02
                  _review_cons_upper  2.399191e-02
                    _review_pros_may  2.485280e-02
          _review_cons_micromanagement  2.542424e-02
                _review_cons_nothing  2.560190e-02
                     _review_cons_hr  2.572595e-02
                _review_cons_horrible  2.624723e-02
                  _review_cons_money  2.669197e-02
                _review_cons_average  2.732930e-02
        _advice_to_management_managers  2.764136e-02
                _review_cons_internal  2.829459e-02
               _review_pros_employees  2.830165e-02
      _advice_to_management_leadership  2.880226e-02
                _review_pros_package  2.907162e-02
                  _review_cons_cares  3.068751e-02
                    _review_cons_20  3.138532e-02
                   _review_cons_need  3.154909e-02
              _review_cons_technology  3.219133e-02
          _advice_to_management_reduce  3.242897e-02
                   _review_cons_made  3.255879e-02
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           _advice_to_management_treat  3.315896e-02
                  _review_cons_values  3.320966e-02
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                _review_cons_expected  3.473952e-02
                _review_pros_work_life  3.491483e-02
                _review_cons_multiple  3.543483e-02
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                     _review_pros_site  3.579552e-02
                     _review_pros_care  3.585433e-02
           _review_pros_good_work_life  3.720657e-02
                       _summary_growth  3.804964e-02
                  _review_cons_minimal  3.823634e-02
                _review_cons_meetings  3.843915e-02
                    _review_pros_learn  3.889315e-02
           _advice_to_management_staff  3.889688e-02
                _review_cons_business  3.896298e-02
                   _review_cons_health  4.032998e-02
        _review_cons_senior_management  4.053186e-02
                    _review_cons_short  4.078345e-02
                      _review_cons_per  4.114178e-02
                _review_pros_diversity  4.182878e-02
         flags_ceo_approval_DISAPPROVE  4.219856e-02
                     _review_pros_well  4.222481e-02
                     _review_cons_call  4.291683e-02
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                           _summary_bad  4.431620e-02
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                  _review_pros_company  4.570402e-02
                    _review_cons_focus  4.637103e-02
                     _review_pros_tech  4.714809e-02
                _review_cons_politics  4.755860e-02
                   _summary_place_work  4.846137e-02
               _review_cons_everything  4.871946e-02
              _review_pros_opportunity  4.911148e-02
                  _review_cons_manager  4.983770e-02
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                    _review_cons_store  5.066417e-02
                _review_cons_behavior  5.067287e-02
                _review_pros_companies  5.096476e-02
                    _review_cons_tasks  5.178098e-02
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                   _review_pros_things  5.229769e-02
          _review_cons_work_environment  5.232928e-02
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                       _review_cons_go  5.391760e-02
        rating_career_opportunities_X3  5.427019e-02
                    _review_pros_years  5.440569e-02
      rating_career_opportunities_X2.5  5.490348e-02
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                    _review_cons_leads  5.544766e-02
                 _review_cons_provided  5.597815e-02
                   _review_cons_person  5.635382e-02
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                   _review_pros_people  5.703594e-02
              _review_pros_good_salary  5.737934e-02
                           _summary_pay  5.771011e-02
                   _review_pros_decent  5.806581e-02
                         _summary_look  5.823425e-02
                 _review_cons_products  5.868187e-02
               _review_pros_everything  5.887596e-02
                        _summary_awful  5.915084e-02
                  _review_cons_respect  5.921010e-02
                    _review_cons_worst  5.922249e-02
        _review_pros_work_life_balance  5.932866e-02
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                     _review_cons_paid  5.981671e-02
                    _summary_underpaid  6.002896e-02
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                   _review_cons_claims  6.074503e-02
             _review_cons_disorganized  6.088377e-02
             _review_pros_life_balance  6.090467e-02
                        _summary_avoid  6.104471e-02
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                        _summary_money  6.210046e-02
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                    _review_pros_teams  6.240121e-02
                      _review_cons_can  6.240232e-02
                    _summary_workplace  6.254688e-02
                _review_cons_corporate  6.260803e-02
                  _review_pros_balance  6.280540e-02
                _review_pros_good_work  6.317586e-02
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                    _review_cons_toxic  6.385280e-02
                _review_pros_technical  6.403091e-02
                 _review_cons_students  6.410669e-02
                _review_pros_generally  6.437085e-02
            _summary_work_life_balance  6.456302e-02
                    _summary_work_life  6.456321e-02
                 _summary_life_balance  6.456341e-02
                 _review_cons_outdated  6.505208e-02
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             _review_cons_inconsistent  6.533095e-02
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                 _review_cons_weekends  6.562648e-02
                  _review_cons_program  6.564528e-02
                    _review_pros_often  6.571132e-02
                     _review_pros_stay  6.579806e-02
                      _review_cons_get  6.589499e-02
          _review_pros_decent_benefits  6.621840e-02
                       _summary_review  6.650827e-02
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        _review_pros_great_people_work  6.791238e-02
         _review_pros_great_place_work  6.797441e-02
               _review_cons_constantly  6.824682e-02
                _review_cons_practices  6.839696e-02
                   _review_cons_owners  6.842833e-02
                         _summary_time  6.858960e-02
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                 _review_pros_students  6.917764e-02
                  _review_cons_product  6.947568e-02
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608                      _review_cons_rate  7.064678e-02
609                _review_pros_good_place  7.085329e-02
610        rating_diversity_inclusion_X2.5  7.086132e-02
611                    _review_cons_enough  7.102776e-02
612         _review_pros_good_pay_benefits  7.117563e-02
613                      _review_pros_week  7.121982e-02
614                         _summary_toxic  7.123391e-02
615               _review_cons_ineffective  7.129197e-02
616            _advice_to_management_teams  7.166872e-02
617                    _review_cons_senior  7.178787e-02
618             _advice_to_management_help  7.189016e-02
619                   _review_pros_awesome  7.216492e-02
620               _review_pros_engineering  7.255406e-02
621                 _review_cons_decisions  7.260758e-02
622                      _review_cons_poor  7.265434e-02
623                  _review_cons_workload  7.267188e-02
624                      _review_pros_much  7.274506e-02
625              _advice_to_management_try  7.320642e-02
626          _review_pros_opportunity_work  7.326940e-02
627                    _review_cons_highly  7.347643e-02
628                         _review_cons_c  7.349677e-02
629                       _review_cons_job  7.368282e-02
630                      _review_cons_want  7.373584e-02
631              _advice_to_management_pay  7.385157e-02
632                    _summary_okay_place  7.410389e-02
633          _advice_to_management_manager  7.427346e-02
634 _advice_to_management_listen_employees  7.429772e-02
635                           _summary_hard  7.437639e-02
636                   _review_cons_low_pay  7.443430e-02
637                _review_cons_leadership  7.453299e-02
638                    _summary_management  7.493980e-02
639                   _review_pros_bonuses  7.499620e-02
640                      _summary_horrible  7.515119e-02
641                  _review_cons_requiring  7.536511e-02
642                      _summary_potential  7.547082e-02
643                           _summary_know  7.547108e-02
644                  _review_pros_learn_lot  7.559476e-02
645                            _summary_job  7.559752e-02
646                    _review_cons_worked  7.614944e-02
647                   _review_cons_project  7.614979e-02
648                      _review_cons_bonus  7.625506e-02
```

```
                _review_cons_questions  7.634270e-02
                _review_pros_paid_time  7.634784e-02
            _advice_to_management_sell  7.669030e-02
                    _review_cons_shift  7.671471e-02
                _review_cons_knowledge  7.678192e-02
                          _summary_get  7.710869e-02
                  _review_pros_learned  7.712199e-02
               _review_cons_especially  7.720211e-02
                      _summary_average  7.722730e-02
            _review_cons_mental_health  7.745182e-02
                      _review_cons_bad  7.784865e-02
                  _review_pros_clients  7.814864e-02
                 _review_pros_personal  7.826010e-02
                     _review_pros_pros  7.826095e-02
                   _review_pros_market  7.828665e-02
              _review_cons_significant  7.865806e-02
               _review_pros_activities  7.887004e-02
                       _summary_better  7.897886e-02
                         _summary_long  7.904433e-02
                      _review_pros_due  7.913416e-02
                 _review_pros_paycheck  7.919807e-02
            _advice_to_management_less  7.923271e-02
       _review_pros_many_opportunities  7.929554e-02
                  _review_cons_another  7.929848e-02
                      _review_cons_now  7.941234e-02
                 _review_cons_software  7.944209e-02
                      _review_cons_pto  7.944853e-02
                   _summary_business  7.964697e-02
                   _review_pros_stable  7.976429e-02
              _review_pros_great_place  7.986489e-02
                _review_pros_excellent  8.022077e-02
                        _summary_dream  8.033100e-02
                   _summary_excellent  8.048956e-02
                     _review_cons_team  8.049759e-02
          _advice_to_management_change  8.049861e-02
                         _summary_nice  8.051847e-02
                     _review_pros_fair  8.081553e-02
                    _review_pros_match  8.094428e-02
                         _summary_much  8.145033e-02
                     _review_cons_feel  8.153839e-02
                  _review_cons_lacking  8.180581e-02
             _review_cons_requirements  8.183582e-02
               _review_cons_long_hours  8.192571e-02
```

```
692                  _review_pros_ability  8.204454e-02
693                          _summary_can  8.211536e-02
694                     _review_pros_part  8.212837e-02
695            _review_cons_high_turnover  8.215530e-02
696                     _review_pros_want  8.245012e-02
697                 _review_cons_policies  8.265081e-02
698                     _review_pros_like  8.265989e-02
699                 _review_pros_location  8.289820e-02
700                          _summary_big  8.301247e-02
701                     _review_cons_know  8.305776e-02
702                  _review_pros_diverse  8.315930e-02
703            _review_pros_international  8.316681e-02
704                     _review_cons_keep  8.320923e-02
705                 _review_pros_generous  8.355100e-02
706              _review_pros_challenging  8.359063e-02
707                     _review_cons_part  8.367287e-02
708                     _review_cons_past  8.372883e-02
709         _advice_to_management_internal  8.376256e-02
710               _review_cons_performance  8.378203e-02
711                     _review_pros_lots  8.385042e-02
712                 _review_pros_managers  8.390720e-02
713                _review_pros_dedicated  8.392530e-02
714                    _review_cons_avoid  8.403594e-02
715            _review_pros_great_company  8.405426e-02
716                  _review_pros_sharing  8.413815e-02
717                          _summary_life  8.452979e-02
718                        _summary_people  8.453179e-02
719             _review_pros_team_members  8.455319e-02
720                  _review_pros_members  8.455364e-02
721                     _review_pros_easy  8.457030e-02
722            _advice_to_management_often  8.462261e-02
723                 _review_pros_knowledge  8.466328e-02
724                   _review_cons_changed  8.479973e-02
725                    _review_pros_travel  8.545628e-02
726                 _review_pros_holidays  8.556901e-02
727                    _review_pros_within  8.557026e-02
728               _review_pros_significant  8.566878e-02
729          _review_pros_health_insurance  8.597655e-02
730                     _review_pros_still  8.605349e-02
731                _review_pros_enjoyable  8.618720e-02
732                _review_pros_potential  8.619601e-02
733                _review_pros_good_team  8.625243e-02
734          _advice_to_management_support  8.630512e-02
```

```
735         _advice_to_management_valuable  8.646341e-02
736                      _review_pros_days  8.648106e-02
737                     _review_cons_blame  8.649768e-02
738                     _review_cons_since  8.666935e-02
739                     _review_pros_space  8.667701e-02
740                    _summary_good_place  8.737840e-02
741                      _review_pros_high  8.741906e-02
742                   _review_cons_reality  8.763033e-02
743                  _review_pros_position  8.765960e-02
744                   _review_pros_enjoyed  8.767454e-02
745            _review_cons_poor_management  8.770359e-02
746                      _review_pros_help  8.826823e-02
747                    _review_cons_making  8.831370e-02
748                  _review_cons_direction  8.841351e-02
749                   _review_cons_properly  8.849297e-02
750                   _review_cons_advance  8.888823e-02
751            _advice_to_management_hiring  8.903278e-02
752                  _review_pros_available  8.918041e-02
753                       _review_pros_fast  8.919412e-02
754                     _review_pros_loved  8.934952e-02
755                  _review_pros_wonderful  8.971634e-02
756           _review_pros_work_environment  8.981842e-02
757                      _review_pros_kind  8.994050e-02
758            _advice_to_management_system  9.000469e-02
759                      _review_cons_help  9.030092e-02
760                   _review_pros_manager  9.030506e-02
761                    _review_cons_skills  9.032130e-02
762                     _review_pros_perks  9.040909e-02
763                   _review_pros_medical  9.071134e-02
764                    _review_cons_people  9.074034e-02
765                 _review_pros_commission  9.075284e-02
766          _advice_to_management_customers  9.091240e-02
767                        _summary_unless  9.091260e-02
768                      _review_pros_team  9.101397e-02
769                  _review_pros_rewarding  9.103832e-02
770                       _review_cons_fast  9.131226e-02
771                    _review_pros_making  9.146491e-02
772               _advice_to_management_day  9.148632e-02
773                   _review_cons_receive  9.159572e-02
774                   _review_pros_overall  9.186458e-02
775                     _review_pros_every  9.214609e-02
776              _advice_to_management_make  9.241397e-02
777                    _review_cons_within  9.244364e-02
```

```
                _review_pros_resources  9.260551e-02
                  _review_cons_breaks  9.280656e-02
                   _review_cons_hires  9.282506e-02
                _review_pros_schedule  9.288482e-02
                  _review_cons_career  9.292259e-02
                    _summary_positive  9.294730e-02
            _advice_to_management_best  9.295222e-02
               _review_pros_place_work  9.318753e-02
                 _review_pros_helpful  9.320569e-02
            _advice_to_management_jobs  9.364294e-02
                          _summary_lot  9.372855e-02
              _review_pros_hardworking  9.373198e-02
     _advice_to_management_individuals  9.382796e-02
                _review_cons_marketing  9.386098e-02
                  _review_cons_family  9.391900e-02
                    _review_pros_make  9.392992e-02
         _review_cons_upper_management  9.401075e-02
                _review_pros_exposure  9.406963e-02
          _advice_to_management_rather  9.412366e-02
                _review_cons_benefits  9.418304e-02
                _review_pros_building  9.431335e-02
                 _review_pros_minimal  9.478558e-02
        _review_pros_opportunity_learn  9.480330e-02
                       _summary_career  9.498023e-02
              _review_pros_nice_office  9.503593e-02
                _review_cons_promotion  9.528746e-02
                  _review_cons_school  9.551673e-02
               _review_pros_healthcare  9.563368e-02
                     _review_pros_paid  9.589365e-02
                      _review_pros_pto  9.628249e-02
                  _review_pros_impact  9.635600e-02
                   _review_cons_month  9.650132e-02
               _review_pros_especially  9.670972e-02
                 _review_cons_leading  9.683171e-02
                 _review_cons_towards  9.683846e-02
                        _summary_solid  9.707809e-02
                  _review_pros_caring  9.712523e-02
                  _review_pros_family  9.716975e-02
                _review_pros_products  9.748923e-02
                          _summary_one  9.757249e-02
            _advice_to_management_team  9.786552e-02
                 _review_cons_revenue  9.792970e-02
                     _review_cons_non  9.817120e-02
```

```
                    _review_cons_care  9.821742e-02
                    _review_pros_rate  9.833673e-02
                        _summary_ever  9.840672e-02
                    _summary_benefits  9.842036e-02
                 _review_cons_bonuses  9.848053e-02
              _review_pros_individuals  9.860998e-02
                _review_cons_promoted  9.879129e-02
                _review_cons_bullying  9.880953e-02
                   _review_cons_owner  9.897200e-02
                 _review_cons_getting  9.908974e-02
                _review_cons_promises  9.920287e-02
                   _review_cons_small  9.926923e-02
                    _review_cons_ever  9.929046e-02
                  _review_cons_mental  9.929124e-02
        _advice_to_management_consider  9.936618e-02
                     _review_cons_pay  9.974837e-02
                _review_pros_provided  9.995573e-02
                  _review_cons_expect  9.998351e-02
                _review_cons_overtime  1.002299e-01
                _review_pros_overtime  1.002467e-01
                _review_cons_promised  1.002603e-01
                    _review_pros_meet  1.003171e-01
                    _review_cons_just  1.004258e-01
               _review_pros_unlimited  1.006301e-01
                    _review_pros_food  1.007095e-01
                _review_pros_everyone  1.007231e-01
                     _review_cons_new  1.008721e-01
                   _review_pros_group  1.010210e-01
                   _review_cons_makes  1.014464e-01
              _review_cons_advancement  1.015411e-01
                    _review_cons_long  1.015708e-01
                 _review_pros_success  1.016224e-01
                    _summary_employee  1.022736e-01
               _review_cons_underpaid  1.023130e-01
                _review_cons_required  1.025127e-01
              _review_cons_departments  1.026143e-01
                 _review_pros_mission  1.027468e-01
                  _review_cons_almost  1.028406e-01
                _review_cons_position  1.028948e-01
                   _review_pros_staff  1.029336e-01
                  _review_pros_friends  1.030483e-01
           _advice_to_management_skills  1.033455e-01
                   _review_cons_gossip  1.034106e-01
```

```
_advice_to_management_numbers  1.037211e-01
  _advice_to_management_based  1.038918e-01
            _review_pros_just  1.045766e-01
        _review_pros_offering  1.047357e-01
         _review_cons_members  1.047482e-01
        _review_cons_industry  1.047837e-01
          _review_cons_always  1.048102e-01
            _review_cons_tech  1.049211e-01
         _review_cons_despite  1.050243e-01
     _review_pros_great_people  1.052328e-01
          _review_cons_system  1.054793e-01
           _review_cons_first  1.056552e-01
        _review_cons_everyone  1.057685e-01
         _review_cons_numbers  1.062618e-01
           _review_cons_never  1.066443e-01
          _review_cons_stress  1.067708e-01
             _review_cons_lot  1.068582e-01
          _review_cons_better  1.070753e-01
          _review_cons_things  1.072162e-01
        _review_cons_turnover  1.076989e-01
            _review_cons_last  1.079892e-01
                _summary_start  1.083859e-01
          _review_cons_hiring  1.087053e-01
        _review_cons_training  1.087327e-01
        _review_cons_employee  1.087398e-01
            _review_pros_cool  1.093207e-01
           _review_cons_given  1.094051e-01
       _review_cons_overworked  1.094403e-01
    _review_pros_flexible_work  1.095013e-01
          _review_pros_amazing  1.096366e-01
         _review_cons_payment  1.098188e-01
            _review_pros_401k  1.101076e-01
     _review_cons_job_security  1.102228e-01
          _review_cons_though  1.106306e-01
        _review_cons_frequent  1.106933e-01
        _review_pros_positive  1.110018e-01
          _review_pros_events  1.110052e-01
       _review_cons_long_term  1.110608e-01
      _review_pros_interesting  1.114581e-01
           _review_pros_tools  1.114733e-01
            _review_cons_hard  1.116030e-01
          _review_pros_skills  1.121916e-01
            _review_cons_many  1.121988e-01
```

```
907                      _review_cons_way  1.122585e-01
908                     _review_cons_much  1.124160e-01
909                     _review_cons_room  1.128758e-01
910                 _review_pros_talented  1.130779e-01
911            _advice_to_management_avoid  1.132602e-01
912                   _review_pros_worked  1.137871e-01
913                 _review_pros_vacation  1.139743e-01
914               _review_pros_people_work  1.140404e-01
915                     _review_cons_also  1.142431e-01
916            _review_cons_care_employees  1.144916e-01
917                  _review_cons_quality  1.147531e-01
918                 _review_pros_employee  1.147799e-01
919                    _review_pros_cares  1.152545e-01
920                     _review_cons_rude  1.153205e-01
921               _review_pros_environment  1.157433e-01
922                 _review_pros_allowing  1.166654e-01
923             _advice_to_management_give  1.168561e-01
924              _review_cons_organization  1.169879e-01
925                  _review_cons_unless  1.178543e-01
926                    _review_cons_takes  1.179804e-01
927                 _review_cons_complete  1.180185e-01
928                    _review_cons_value  1.187309e-01
929                    _review_cons_weeks  1.192222e-01
930               _review_pros_development  1.195836e-01
931                  _review_cons_workers  1.200198e-01
932                     _review_cons_laid  1.206498e-01
933            _review_pros_great_benefits  1.211054e-01
934                 _review_pros_benefits  1.216685e-01
935                    _review_pros_areas  1.217565e-01
936                 _review_cons_security  1.222401e-01
937                  _review_cons_quickly  1.233080e-01
938                     _review_cons_sick  1.238653e-01
939              _advice_to_management_due  1.244717e-01
940                    _review_cons_fired  1.251606e-01
941                         _review_cons_4  1.253352e-01
942                  _review_cons_clients  1.256475e-01
943                      _review_cons_may  1.261787e-01
944            _advice_to_management_needs  1.262549e-01
945                 _review_cons_accounts  1.266162e-01
946                  _review_pros_culture  1.289440e-01
947                    _review_cons_every  1.294791e-01
948               _review_pros_atmosphere  1.303142e-01
949    _review_pros_learning_opportunities  1.306017e-01
```

```
950                    _review_cons_others  1.309052e-01
951                       _review_pros_one  1.309718e-01
952                     _review_cons_place  1.316227e-01
953                     _review_cons_rules  1.317634e-01
954                     _review_cons_daily  1.332092e-01
955                      _review_cons_zero  1.368071e-01
956                     _review_pros_hours  1.393675e-01
957                  _review_cons_holidays  1.397161e-01
958                  _review_cons_managers  1.647587e-01
```

Elastic Net

```
results = list()
for(a in seq(0,1,0.05)){
  model.length.elastic <- cv.glmnet(X_train_processed, yq_train, family = binomial, alpha = a
  results[[as.character(a)]] <- min(model.length.elastic$cvm)
}

# find best alpha
best_alpha <- names(which.min(unlist(results)))
print(paste("The best Alpha is:", best_alpha))
```

```
[1] "The best Alpha is: 0.9"
```

```
model.length.elastic <- cv.glmnet(X_train_processed, yq_train, family = binomial, alpha = bes

model.length.elastic_coef <- predict(model.length.elastic, s = model.length.elastic$lambda.mi

model.length.elastic_coefs <- as.matrix(model.length.elastic_coef)
model.length.elastic_vars <- data.frame(
  Variable = rownames(model.length.elastic_coefs),
  Coefficient = as.numeric(model.length.elastic_coefs)
) %>% filter(Coefficient != 0 & Variable != "(Intercept)") %>%
  arrange(Coefficient)


model.quit.elastic_vars
```

```
                        Variable  Coefficient
```

```
1              _review_pros_covid -1.790353368
2              _review_cons_covid -1.474711075
3               _review_pros_name -1.437811527
4         covid_related_review_X1 -0.914125729
5              _review_cons_still -0.293124400
6    _advice_to_management_ensure -0.033979811
7 _advice_to_management_processes -0.004479564
8            career_missing_TRUE.  0.328456874
```